AI4InSync

Optimizing Southern Ocean observing systems using AI.

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ANTARCTICA INSYNC

AI4InSync

A collaborative workshop to support observation system design with machine learning methods.

Workshop Outcomes:

(a) Identify 1-3 scientific questions
(b) Identify potential working groups

InSync Webinar // 8-10am // 22 May

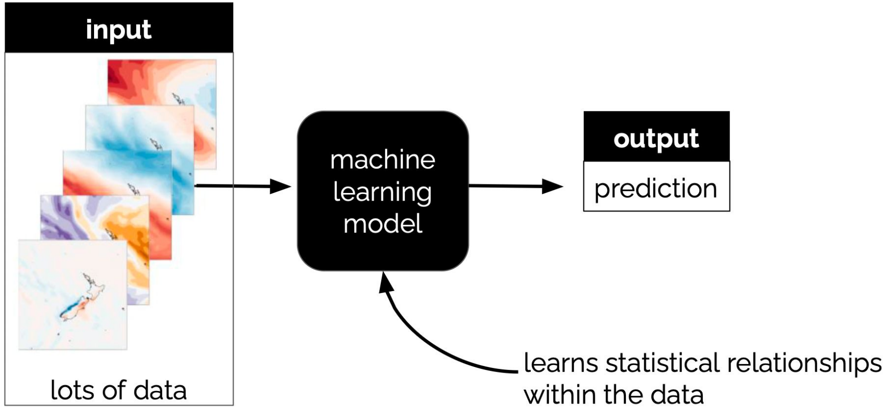
Session 1	Session 2	Session 3	Session 4
Motivation & Intro to ML	**ML Tutorial**	**State-of-the-Art: ML in Climate Science**	**Discussion:** Identify science questions Define potential working groups

AI4InSync

Current state-of-the-art for ML in Climate Science.

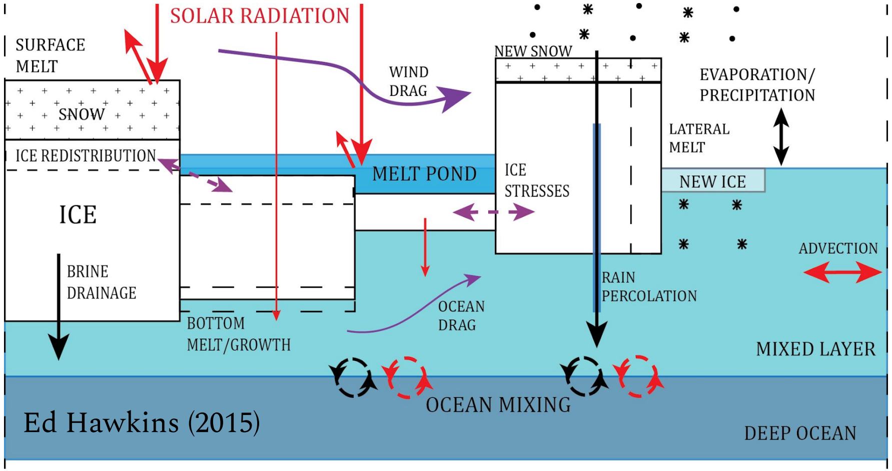
ML models learn statistical relationships in data, whereas numerical models are based on prescribed physics.

Machine Learning Models



Rampal et al. (2024)

Numerical Models



ML models *learn from data* to recognize patterns and make predictions based on those patterns.

Numerical models are *prescribed dynamical equations* and predict based on physics.

Machine Learning Models

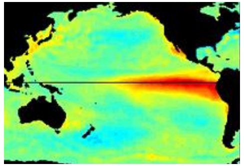
- Lack physical process knowledge
  - *there are efforts to improve this*
- The 'black box' is difficult to interpret
  - *there are efforts to improve this*
- Rely on statistical interactions in data
- Computationally efficient

Numerical Models

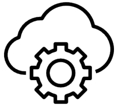
- Rooted in physics
- Interpretable
- Rely on parameterizations for small-scale processes
  *there are efforts to improve this numerically **and with ML***
- Computationally inefficient

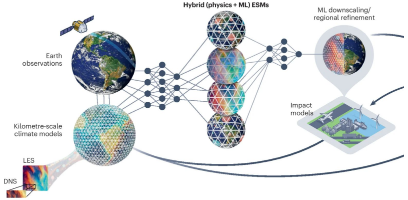
There are an number of applications for ML in Earth and climate sciences, including (but not limited to)...

Sources of predictability for modes of climate variability

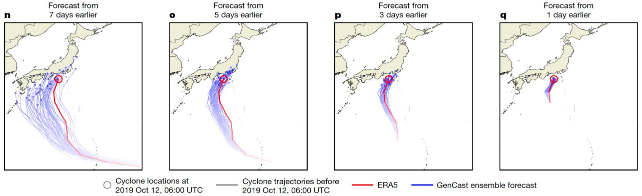


Earth System Modeling



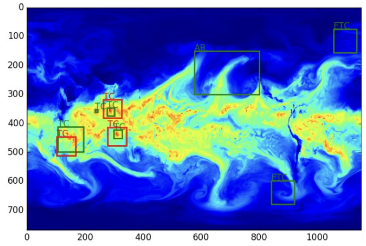


Extreme weather and climate prediction

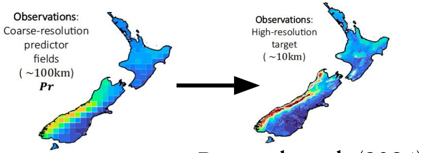


Price et al. (2024)

Feature Detection

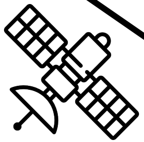


Racach et al. (2017)



Rampal et al. (2024)

Satellite Data & Observations



But incorporating ML into Earth and climate science still has its challenges...

Challenges
Data Availability
Robustness on out-of-distribution samples
Interpretability
Physical Inconsistency
Uncertainty Quantification

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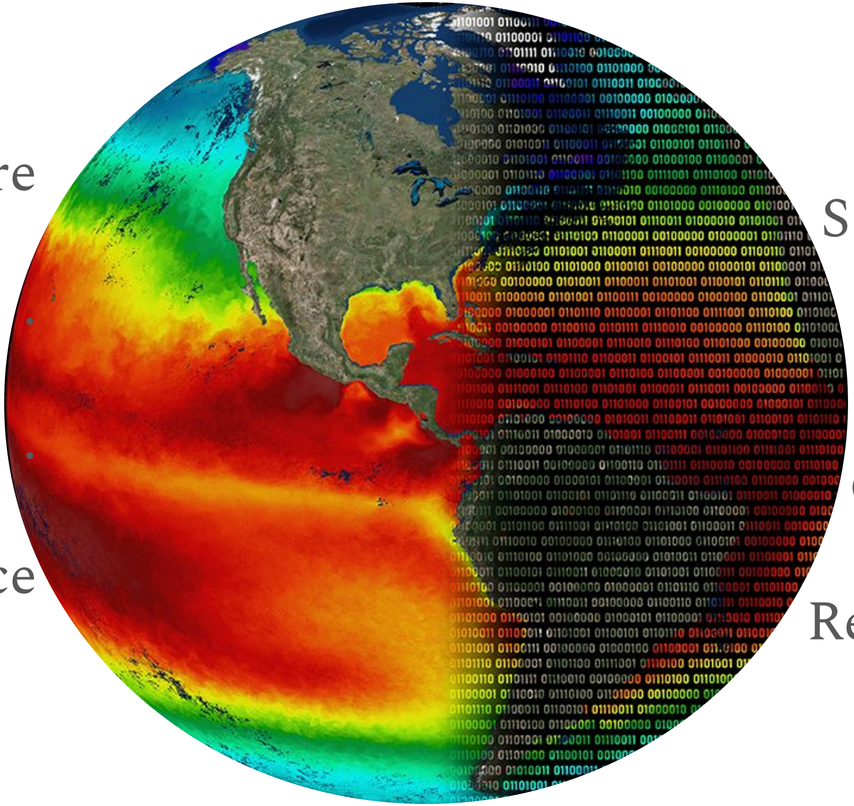
Data Availability: it is impossible to measure everything →
Climate Model Output can be used to train a ML model.



ML ♥ DATA

Image: Courtesy of Kate Culpepper with design elements provided by Esri, HERE, Garmin, FAO, NOAA, USGS, EPA.

But incorporating ML into Earth and climate science still has its challenges...

Challenges
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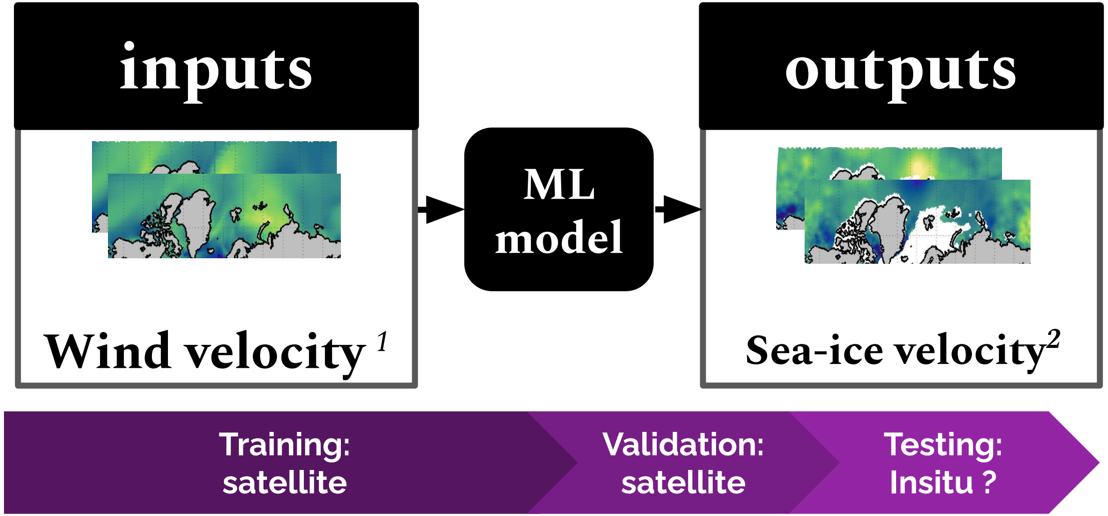
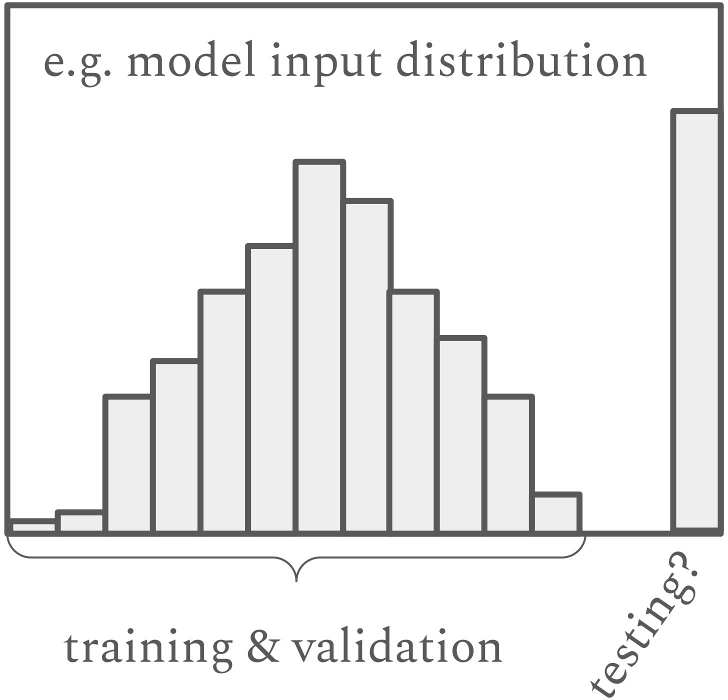
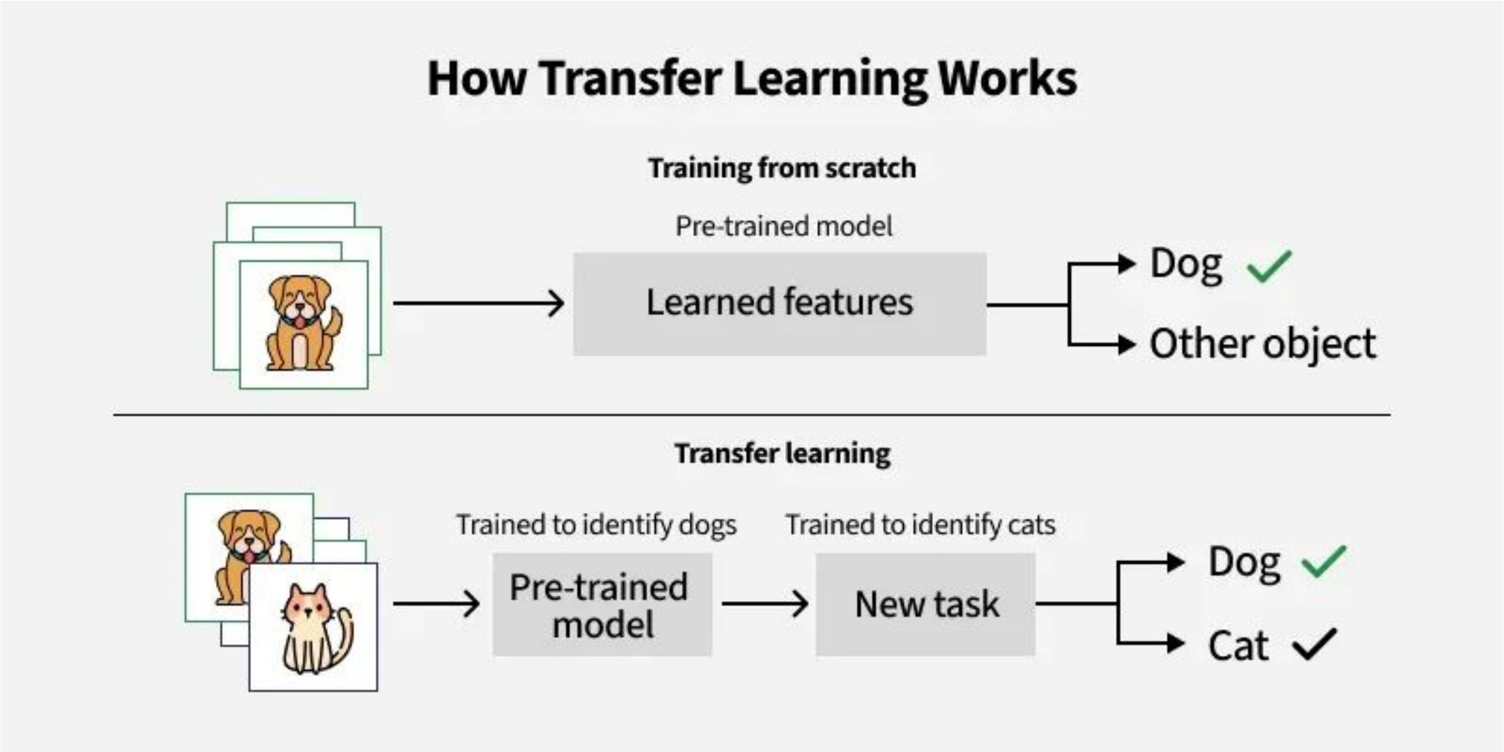
Robustness on Out-of-Distribution Samples:
ML models often have difficulty generalizing.





Robustness on Out-of-Distribution Samples →



Transfer Learning uses a pre-trained model and new data to learn more information (i.e. further improve loss function).

But incorporating ML into Earth and climate science still has its challenges...

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Interpretability:
ML models can learn the right thing for the wrong reasons.

Clever Hans: the horse that can do math!

Or can he....

Interpretability → eXplainable AI (XAI)

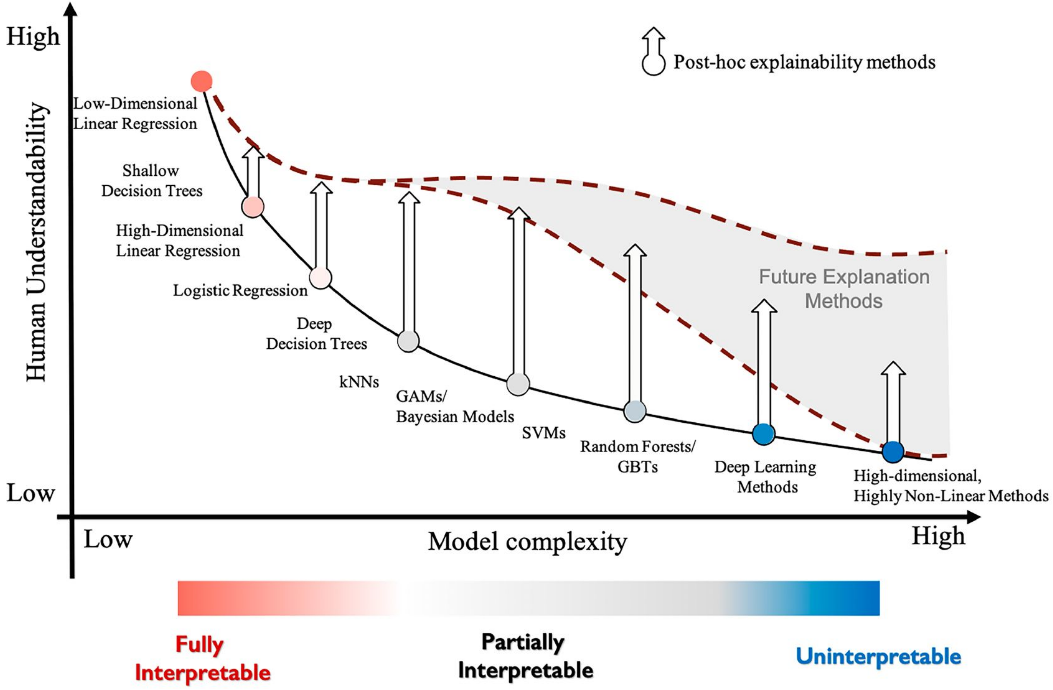


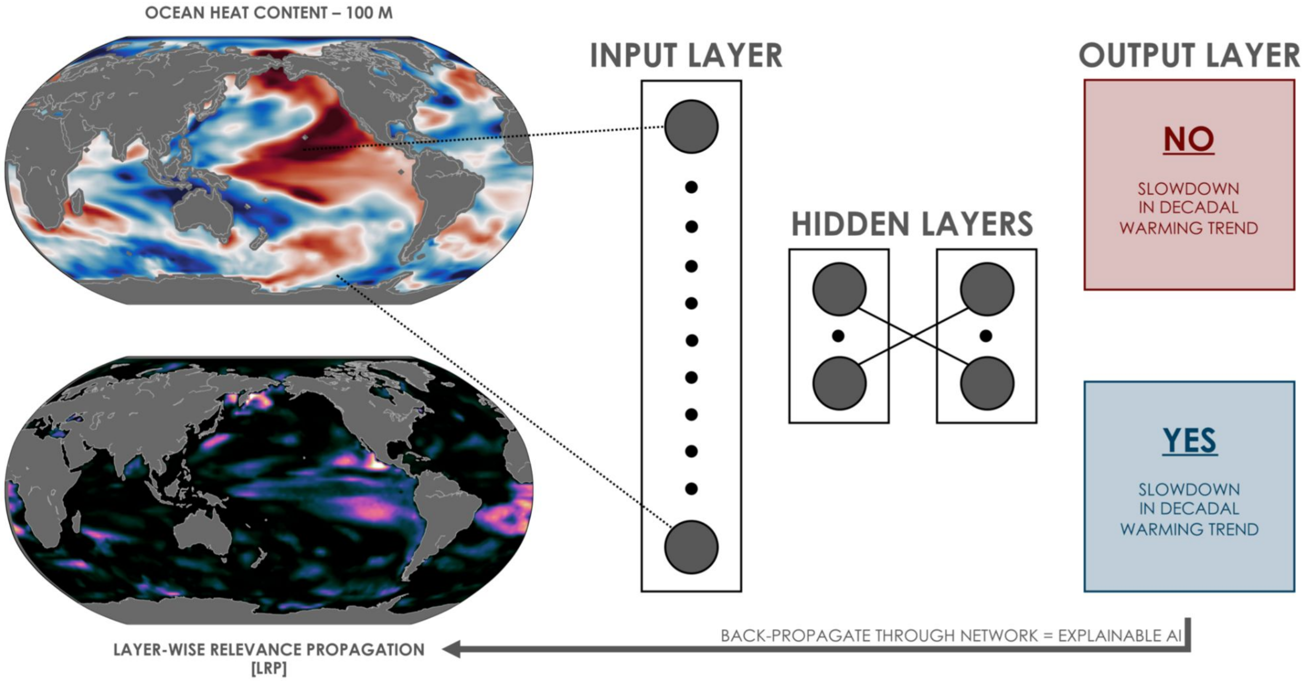
FIG. 1. Illustration of the relationship between understandability and model complexity. Fully interpretable models have high intrinsic understandability, while partially interpretable or simpler black box models have the most to gain from explainability methods. With increased dimensionality and nonlinearity, explainability methods can improve understanding. Still, there is considerable uncertainty about the ability of future explanation methods to improve the understandability of high-dimensional, highly nonlinear methods.

Models that are more complex tend to be less interpretable.

Flora et al. (2024)

Interpretability → eXplainable AI (XAI)

A neural network predicts slowdowns in decadal warming trends from maps of ocean heat content. ***eXplainable AI shows where the inputs were relevant.***



Labe & Barnes (2022)

But incorporating ML into Earth and climate science still has its challenges...

Challenges
Data Availability
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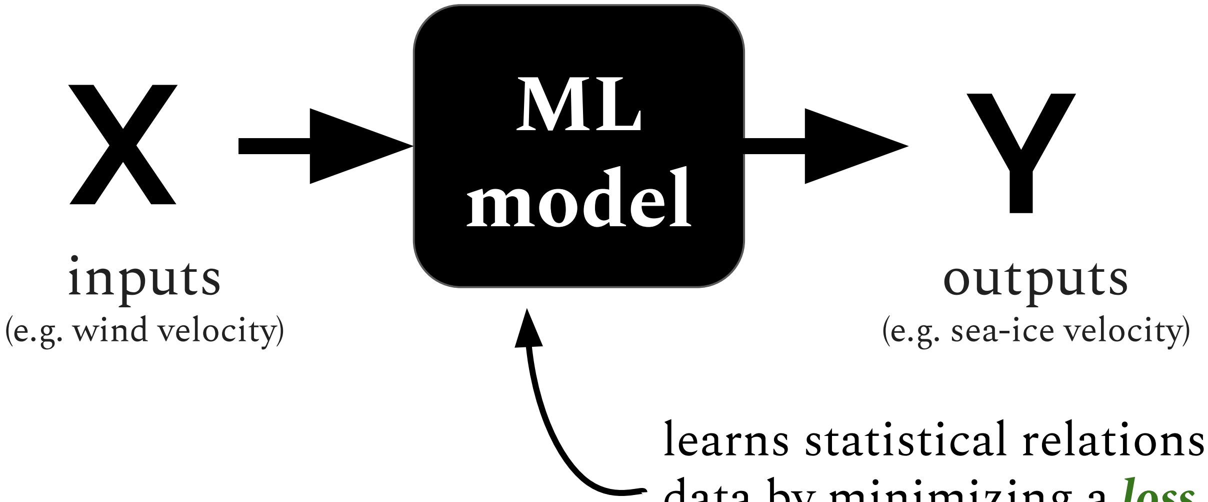
Physical Inconsistency
Uncertainty Quantification

Physical Inconsistency → Physics-Informed ML



learns statistical relationships within the data by minimizing a *loss function*

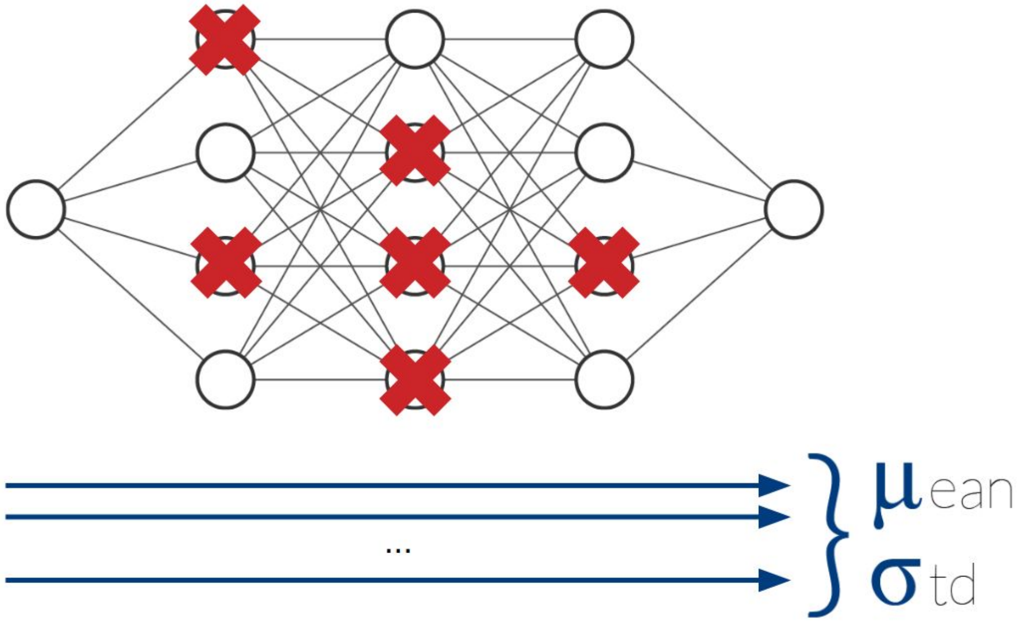
We can apply a custom loss function that includes information about physics (i.e. deviation from conservation of mass, etc.)

Liu et al (2024)

But incorporating ML into Earth and climate science still has its challenges...

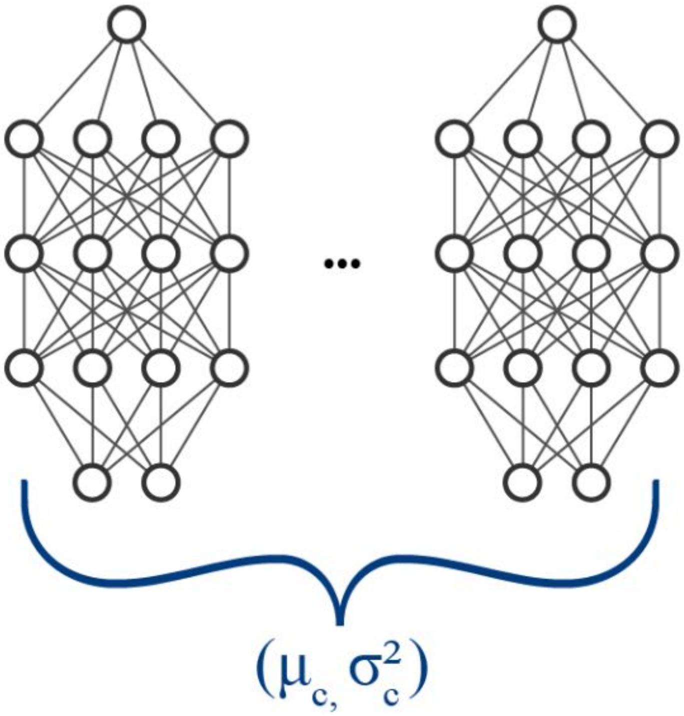
Challenges
Data Availability
Robustness on out-of-distribution samples
Interpretability
Physical Inconsistency
Uncertainty Quantification

Uncertainty Quantification → Monte Carlo Dropout



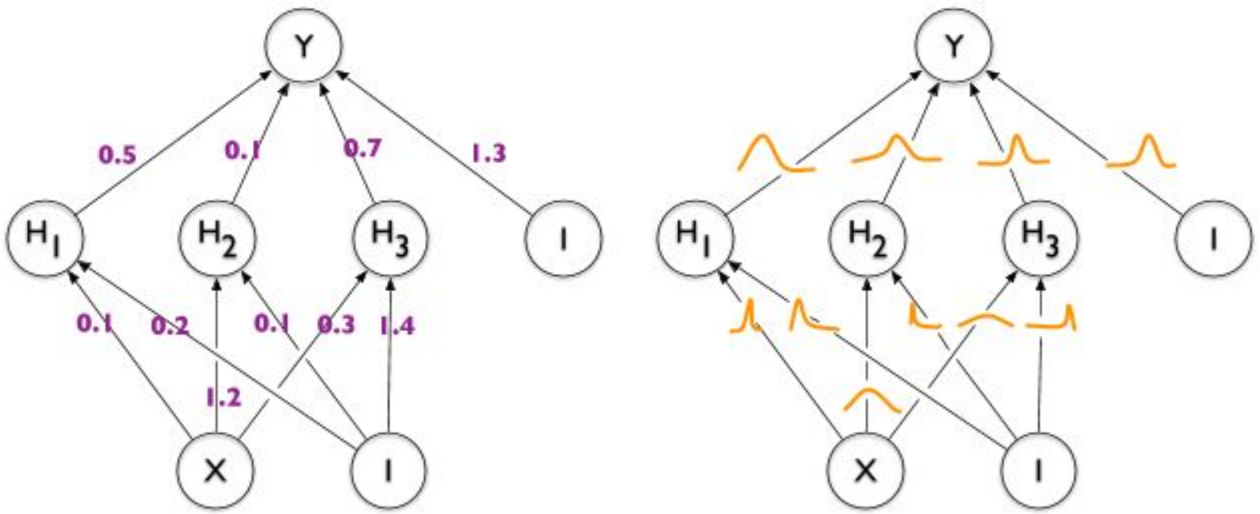
During Monte Carlo Dropout ***nodes are randomly disabled during training*** so that for each epoch a random subset of weights and biases is evaluated and adjusted.

Uncertainty Quantification → Ensemble Averaging



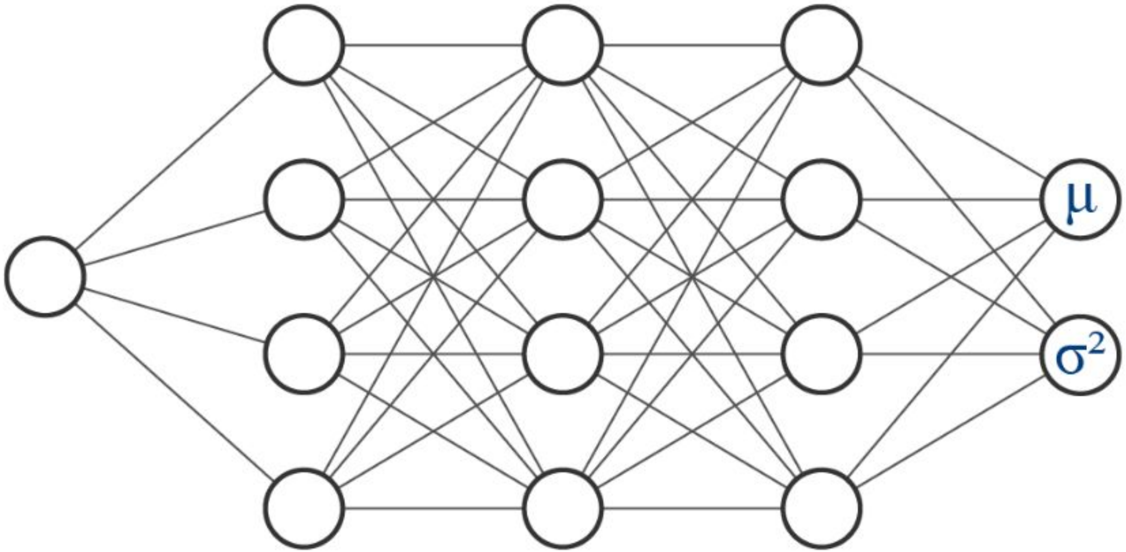
Train an ensemble of neural networks with different initializations.

Uncertainty Quantification →
Bayesian Neural Networks (BNNs)



In BNNs, the weights are distributions not deterministic values.

Uncertainty Quantification → Distributional Parameter Estimation



A neural network predicts the mean and variance, rather than a deterministic value.

Incorporating Uncertainty into a regression neural network enables identification of decadal-state-dependent predictaility in CESM2.

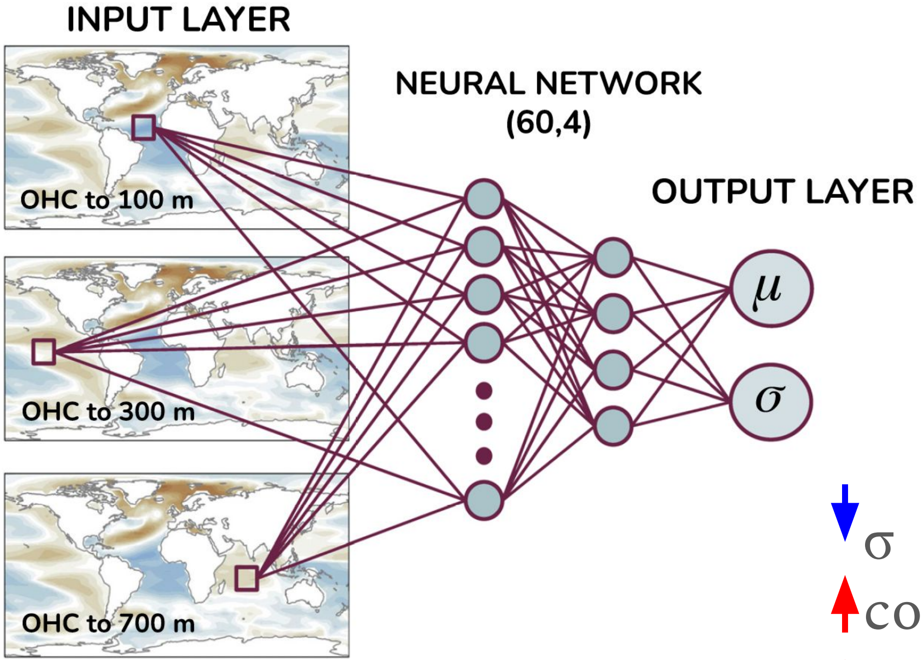
Gordon and Barnes (2022)

Inputs:

- Layers of OHC
- Different layers contain different information about climate variability

Output:

- Atlantic Ocean SST anomaly
- Gridded
- 60-month average
- 1-5 and 3-7 years lead time



Not penalized for high error predictions if a high σ is recognized

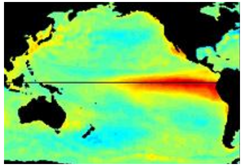
↓ σ
↑ confidence
↑ predictability

But incorporating ML into Earth and climate science still has its challenges...

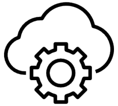
Challenges	Solutions
Data Availability	Climate model output
Robustness on out-of-distribution samples	Transfer learning
Interpretability	eXplainable AI
Physical Inconsistency	Physics-informed ML
Uncertainty Quantification	BNNs, Ensembles, Dropout, etc.

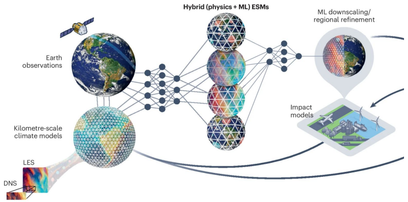
There are an number of applications for ML in Earth and climate sciences, including (but not limited to)...

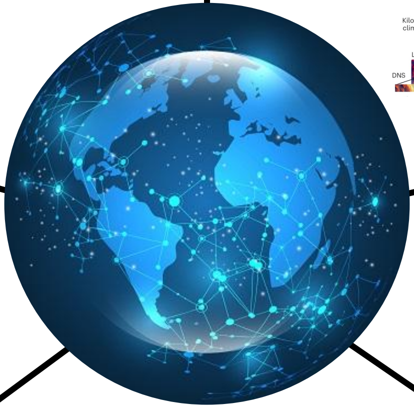
Sources of predictability for modes of climate variability



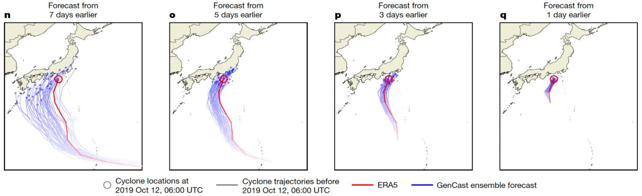
Earth System Modeling





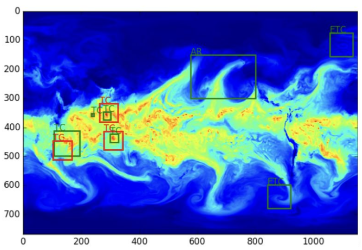


Extreme weather and climate prediction

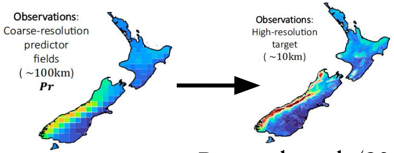


Price et al. (2024)

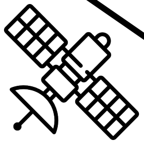
Feature Detection



Racah et al. (2017)

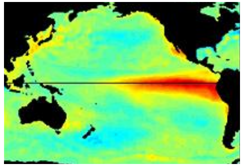


Rampal et al. (2024)



Satellite Data & Observations

Climate Regimes and Drivers



Climate Modes

Causal analysis

Clustering

Forecasting

Short-term weather forecasting

Extreme Weather Events

Climate Forecasting

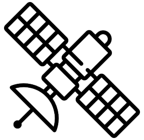
Satellite Data & Observations

Interpolation

Global Reconstruction

Variable Representation

Downscaling



Merging Satellite Data

Synthetic data

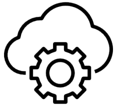
Sensor placement

Feature Detection

Extremes

Pattern Recognition

Earth System Modeling



Emulators

Model Calibration and Validation

Equation Discovery

Uncertainty Quantification

Sources of uncertainty

Parameterizations

Bias Correction

Data Assimilation

Climate Model analysis and benchmarking

Forecasting: Diffusion Models

Price et al. (2024)

Diffusion model

Article

Probabilistic weather forecasting with machine learning

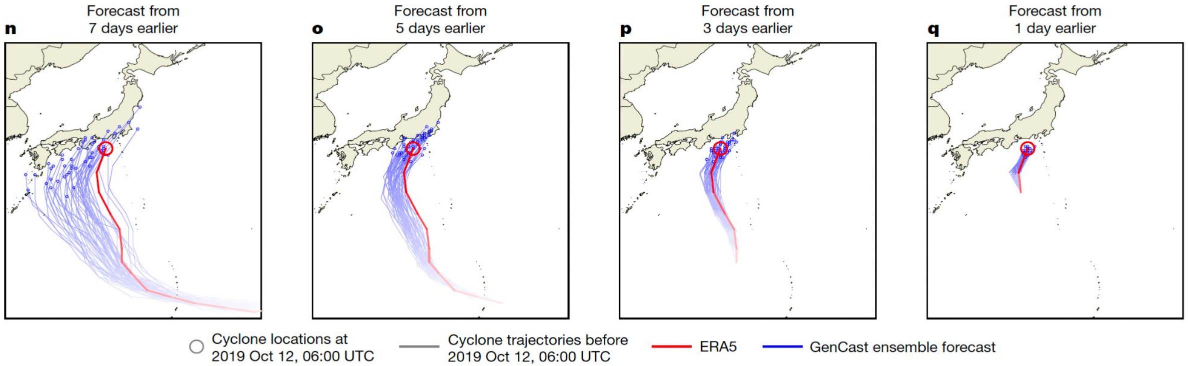
https://doi.org/10.1038/s41586-024-08252-9

Received: 30 April 2024

Accepted: 18 October 2024

Published online: 4 December 2024

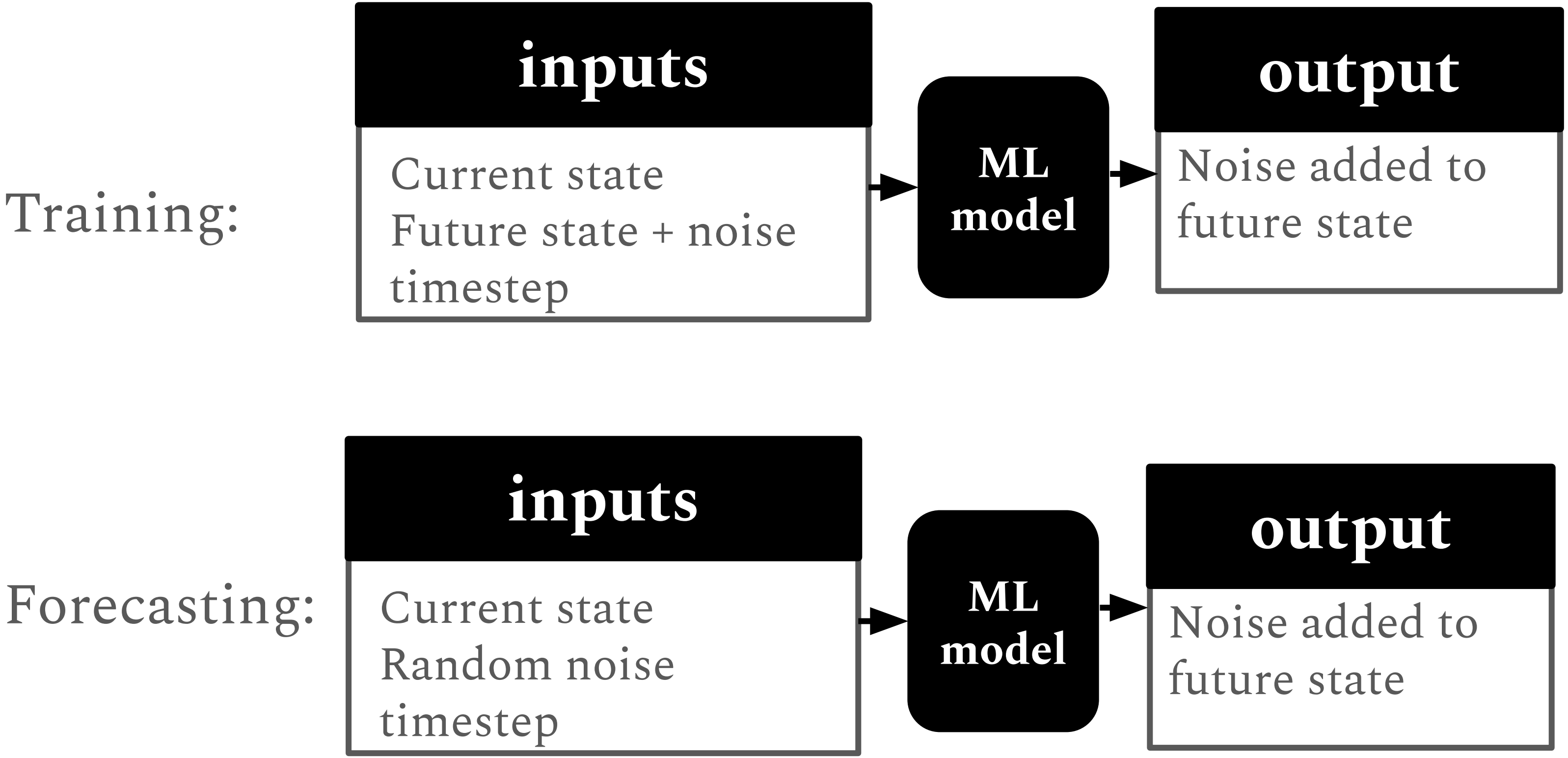
Ilan Price[1,2], Alvaro Sanchez-Gonzalez[1,2], Ferran Alet[1,2], Tom R. Andersson[1,2], Andrew El-Kadi[1], Dominic Masters[1], Timo Ewalds[1], Jacklynn Stott[1], Shakir Mohamed[1], Peter Battaglia[1], Remi Lam[1] & Matthew Willson[1]



"This work presents GenCast, the first MLWP method, to our knowledge, that significantly outperforms the top operational ensemble NWP model, ENS."

Forecasting: Diffusion Models

Price et al. (2024)



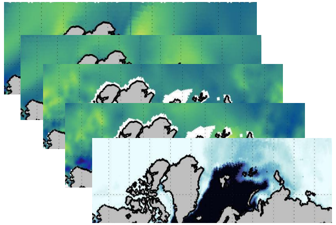
Future state = random noise (input) - noise added to future state (learned)

Forecasting: Daily, SIU & SIV

Hoffman et al. (2023)

A convolutional neural network (CNN) trained on satellite and reanalysis data makes skillful one-day predictions of Arctic sea ice motion.

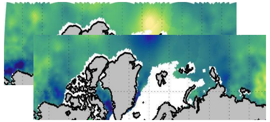
inputs



Present-day wind velocity [1]
Previous-day sea-ice velocity [2]
Previous-day sea-ice concentration [3]

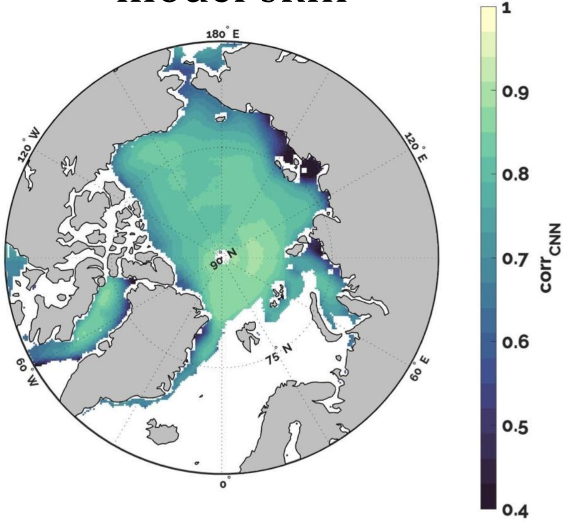
→ **ML model** →

outputs



Present-day sea-ice velocity

model skill



+ XAI for variable importance
+ surrogate model

Model Inputs: [1]*JRA55;* [2]*Polar Pathfinder;* [3]*Nimbus-7*

Seasonal to decadal (S2D) probabilistic forecasts of Arctic September sea ice extent with statistical models.

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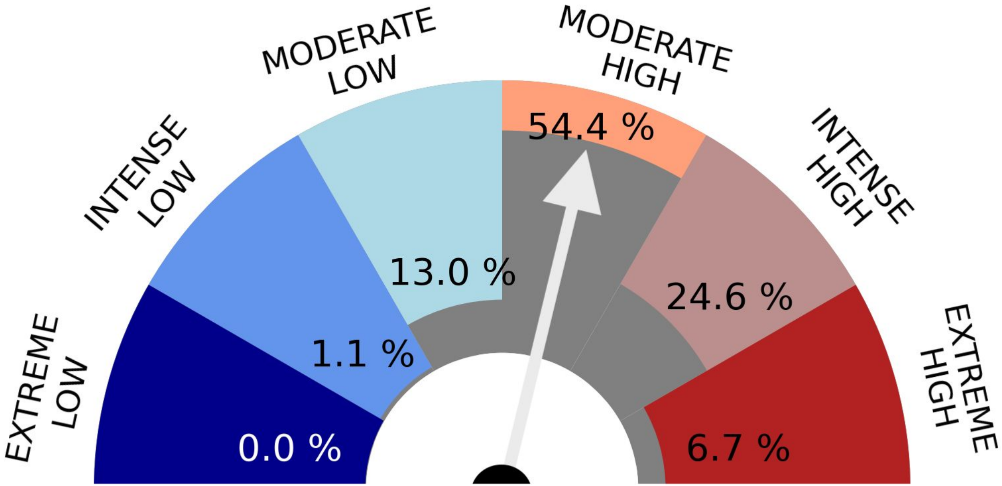
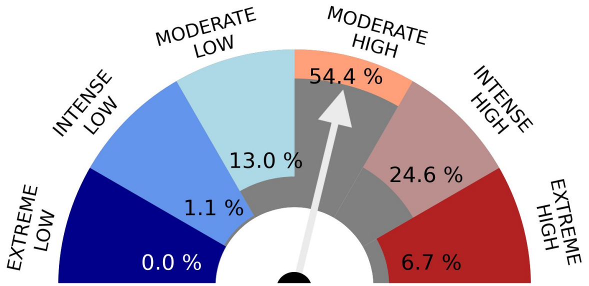
Lauren Hoffman, François Massonnet, Annelies Sticker
Earth and Life Institute (ELI), Earth and Climate, Université catholique de Louvain

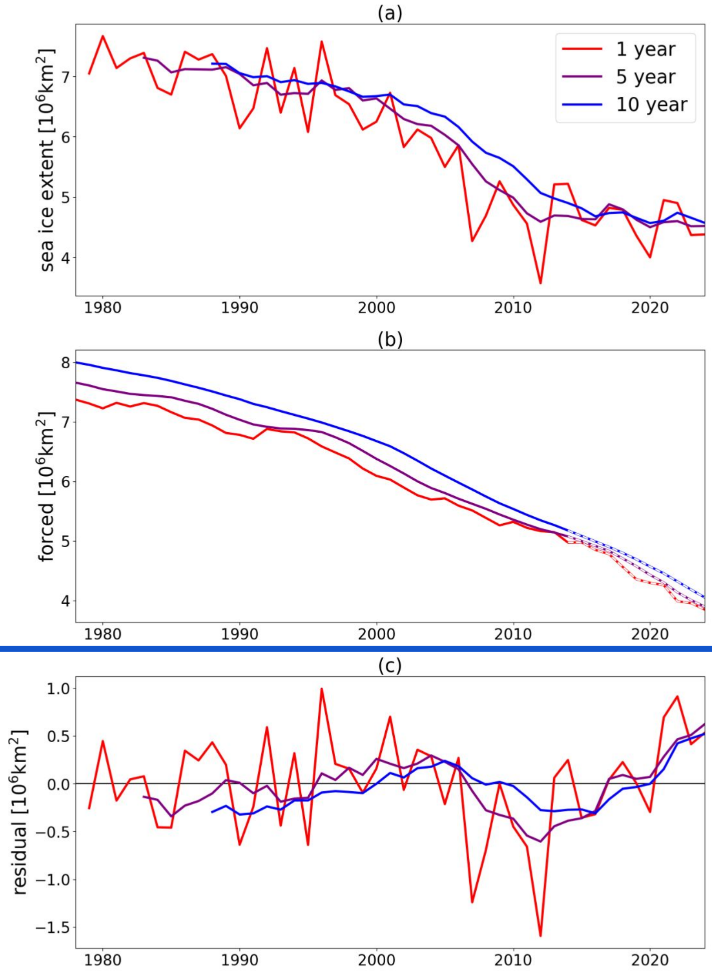


Our models predicts an 86% chance of anomalously high September SIE for 2024–2028.

Our statistical models predict the *internal variability* of September Arctic SIE.

There is 86% chance of anomalously high September SIE for 2024–2028.





SIE

=

Forced

(CMIP6 mean)

+

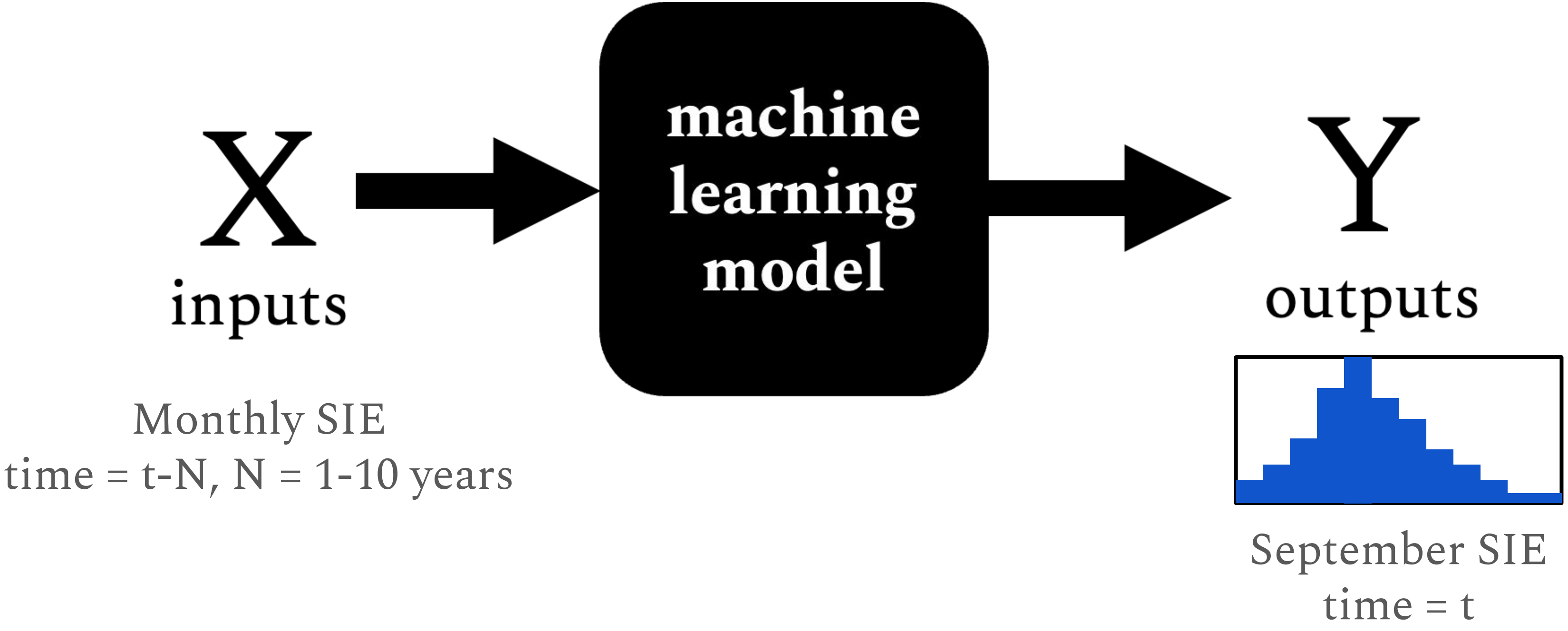
Residual

(internal variability)

SIE Internal Variability

Hoffman et al. (*submitted, 2025*)

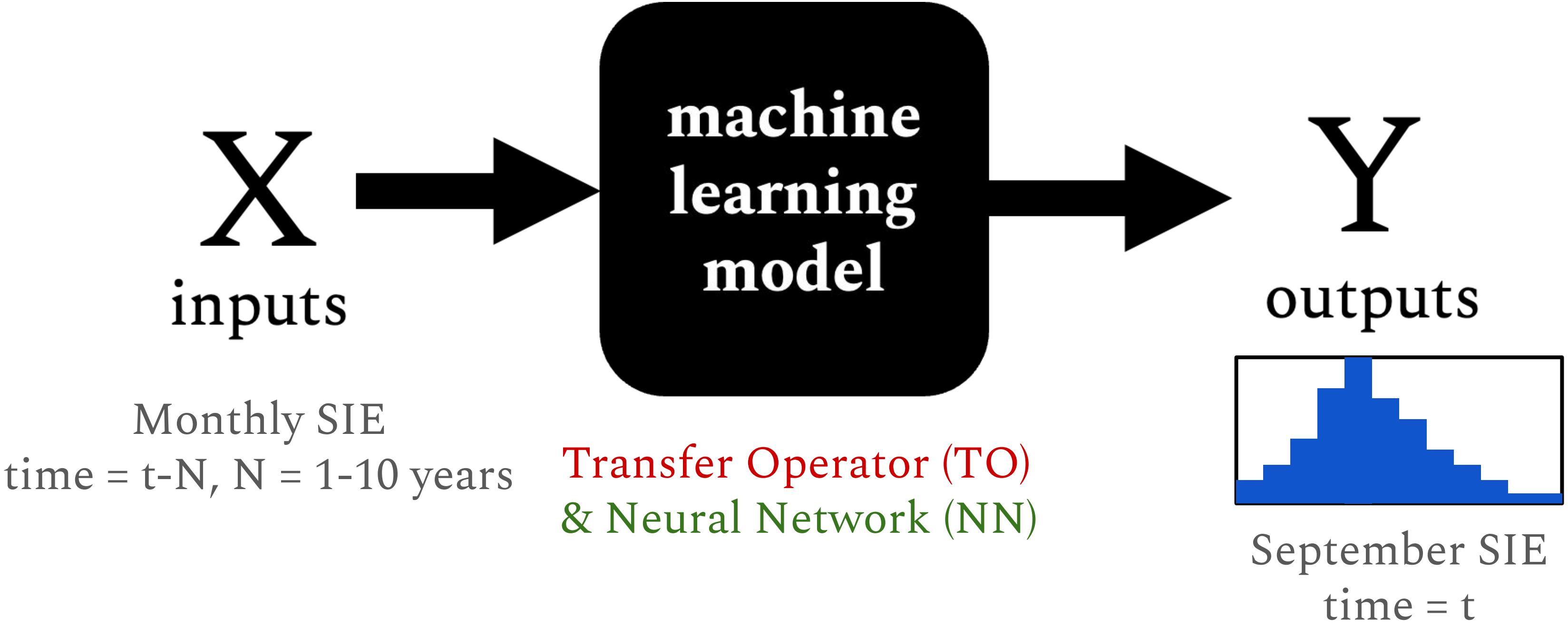
Machine learning models are trained to predict *state transitions* in the internal variability of Arctic SIE using CMIP6 outputs.



Model Architectures

Hoffman et al. (*submitted, 2025*)

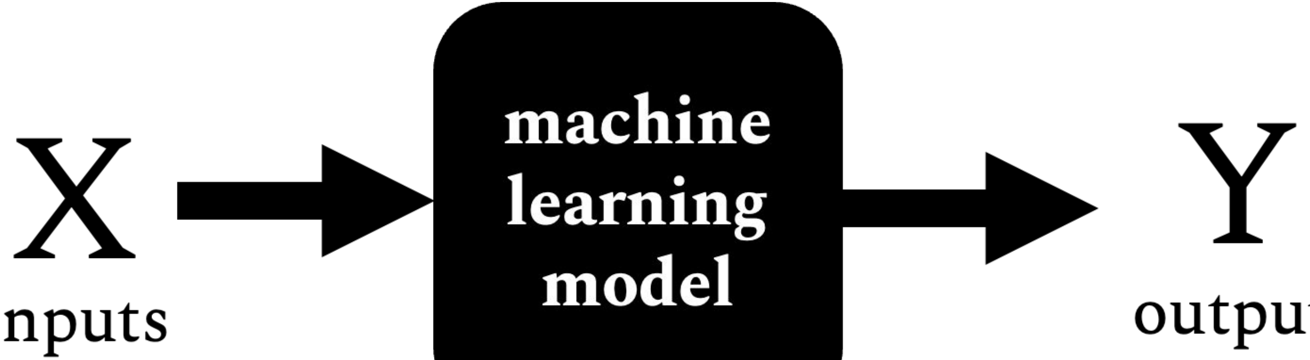
We build *transfer operators and neural networks* to predict state transitions of sea ice extent (SIE).

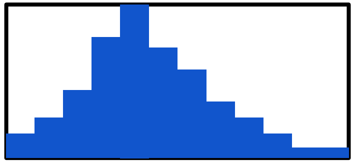


Model Architectures

Hoffman et al. (*submitted, 2025*)

The neural network has an additional input parameter: the *area of thick ice (SIAt).*





TO: Monthly SIE
NN: Monthly SIE + **SIAt**
time = t-N, N=1-10 years

Transfer Operator (TO)
& Neural Network (NN)

September SIE
time = t

Model Architectures

Hoffman et al. (*submitted, 2025*)

The *area of thick ice (SIAt)* is important for sea ice predictability.

There is a relationship between ***melt probability and sea ice thickness***.

Goosse et al. (2009)

"Ice area anomaly in September is potentially predictable up to 6 months in advance using ***area covered by ice thicker than a critical thickness*** lying between 0.9 and 1.5m."

Chevallier and Salas-Mélia (2012)

"***Thickness observations collected after melt onset*** are particularly critical for summer Arctic sea ice predictions."

Bushuk et al. (2020)

The *area of thick ice (SIAt)* is important for sea ice predictability.

How does additional information about the *area of thick ice (SIAt)* impact the *skill* and *behavior* of our model?

thickness.

Goosse et al. (2009)

“Ice area anomaly in

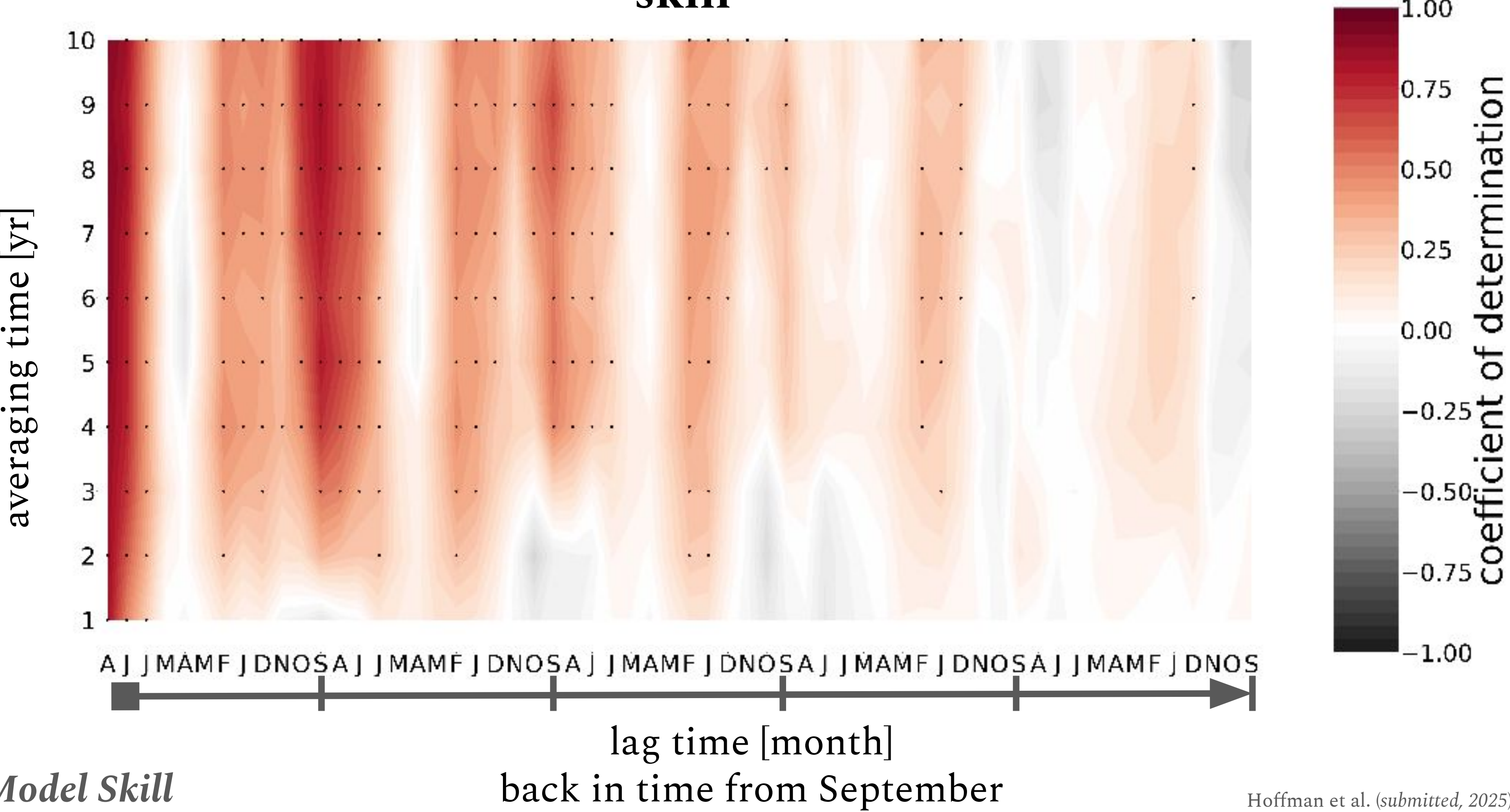
thicker than a critical thickness lying between 0.9 and 1.5m.”

Chevallier and Salas-Mélia (2012)

predictions.”

Bushuk et al. (2020)

skill



Model Skill

Hoffman et al. (*submitted, 2025*)

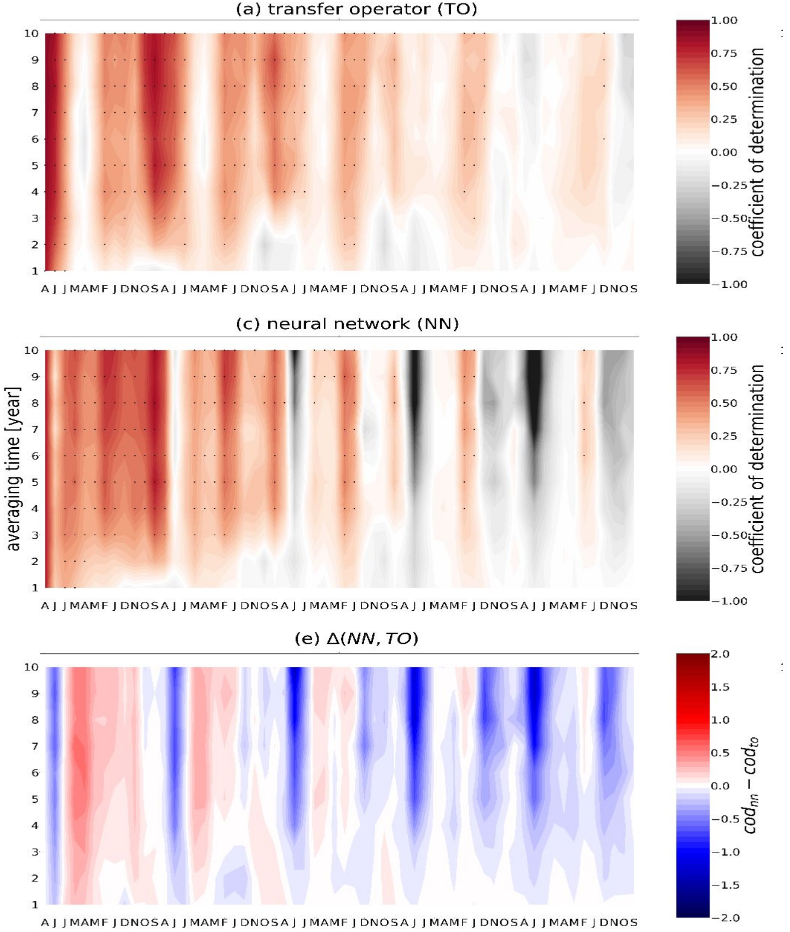
Both models perform reasonably well on observations...

Transfer Operator (TO)

Neural Network (NN)

Δ NN-TO

skill



Model Skill

Hoffman et al. (*submitted, 2025*)

Both models perform reasonably well on observations…

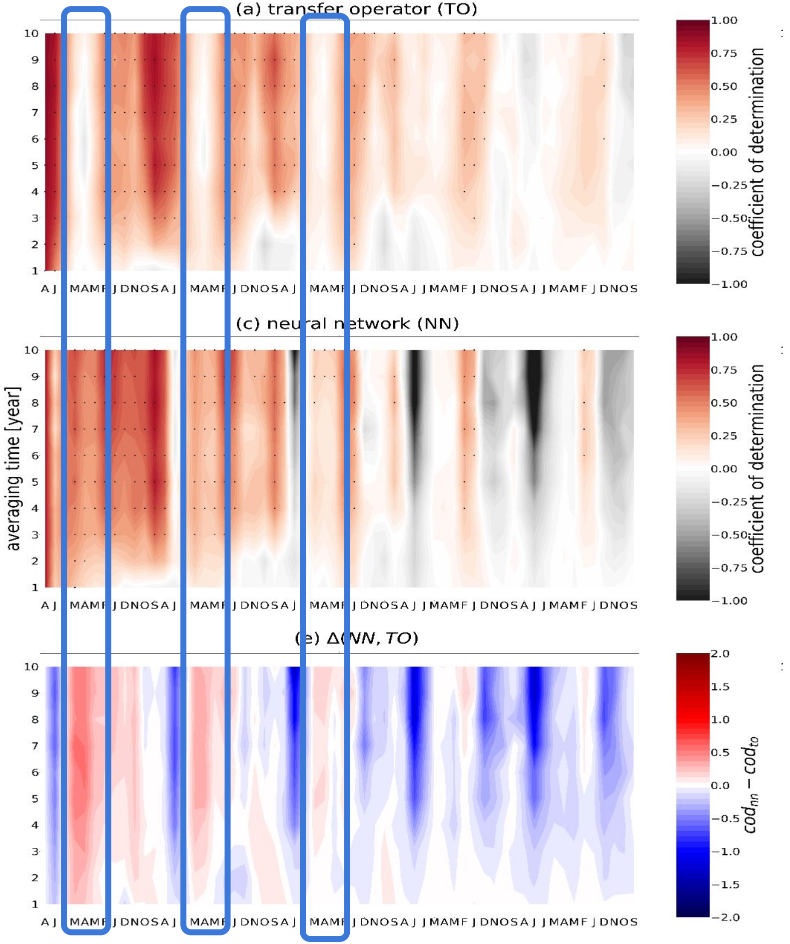
TO performances exhibit the characteristic *spring predictability barrier,* but not the NN.

skill

Transfer Operator (TO)

Neural Network (NN)

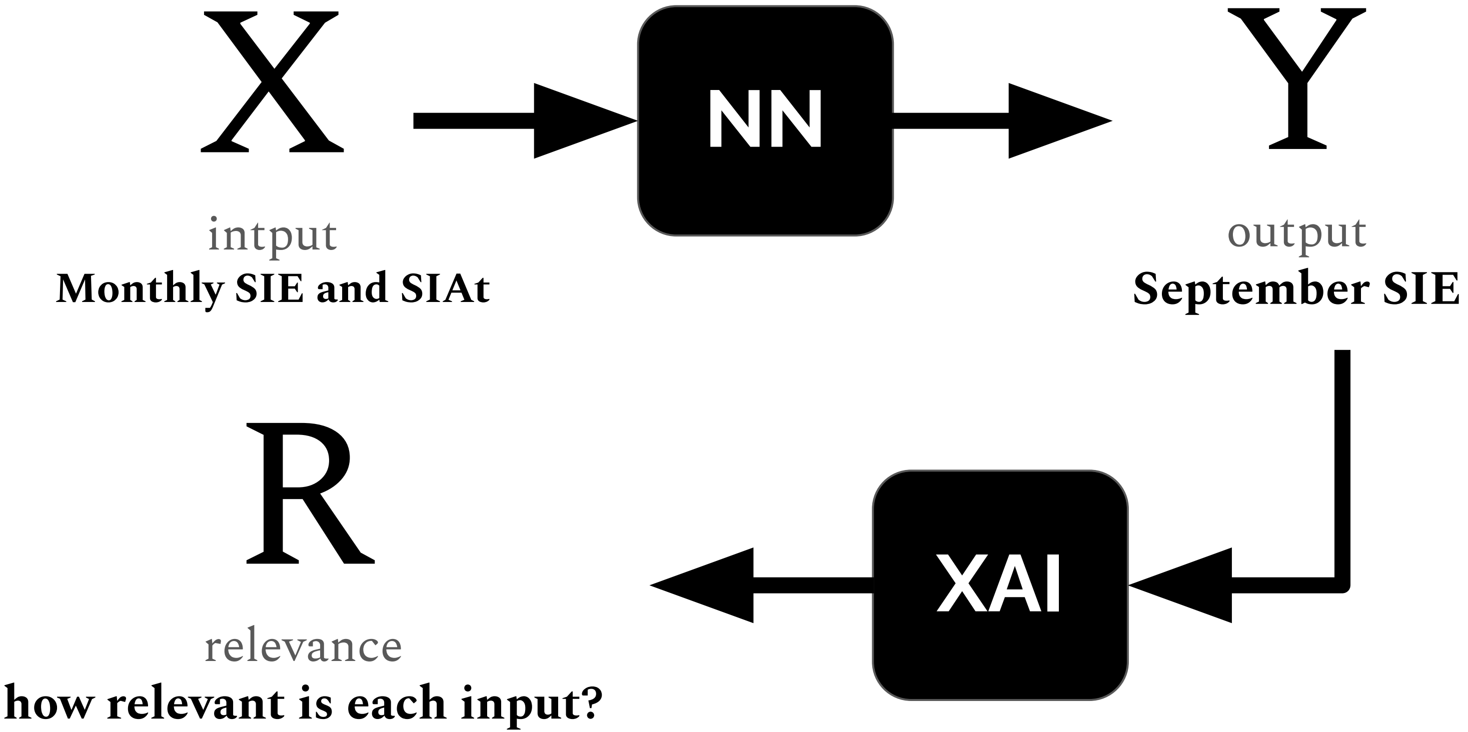
Δ NN-TO



Model Skill

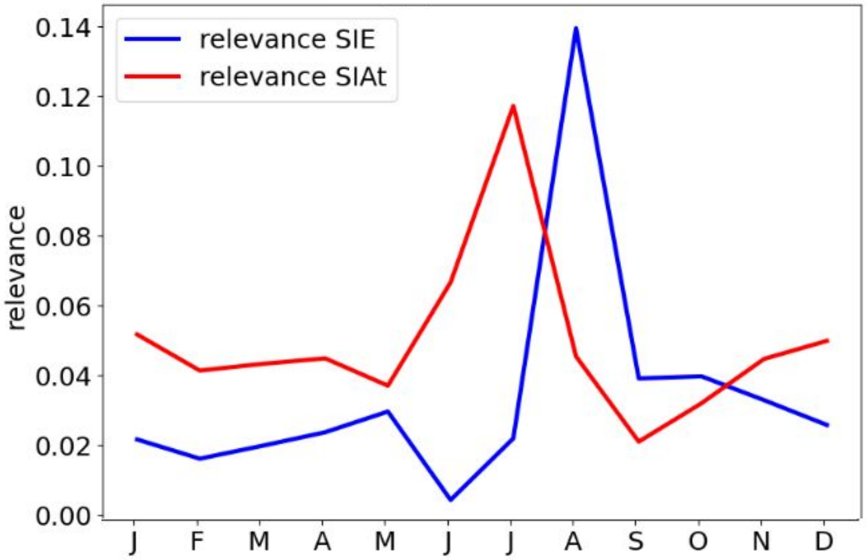
Hoffman et al. (*submitted, 2025*)

eXplainable AI shows how relevant each of the inputs is for the neural network in predicting September SIE.



Model Behavior

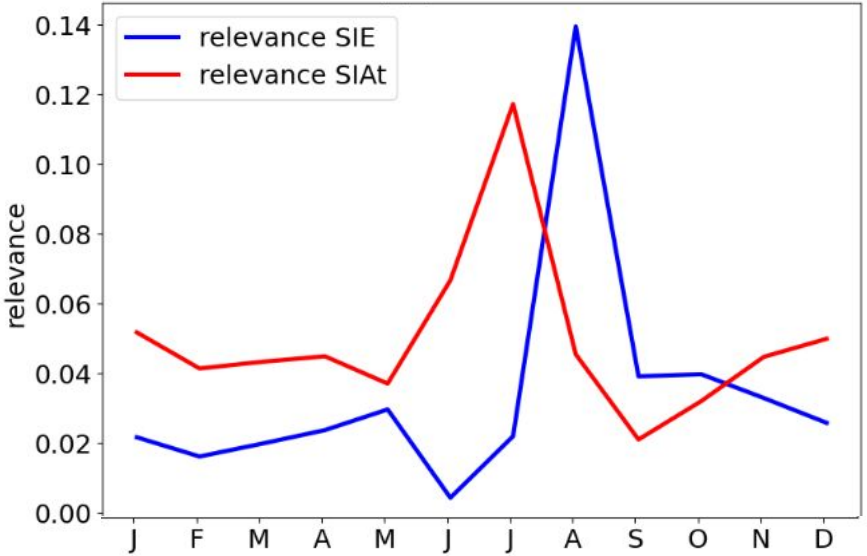
eXplainable AI perturbation studies show the relevance of the inputs for predicting September SIE.



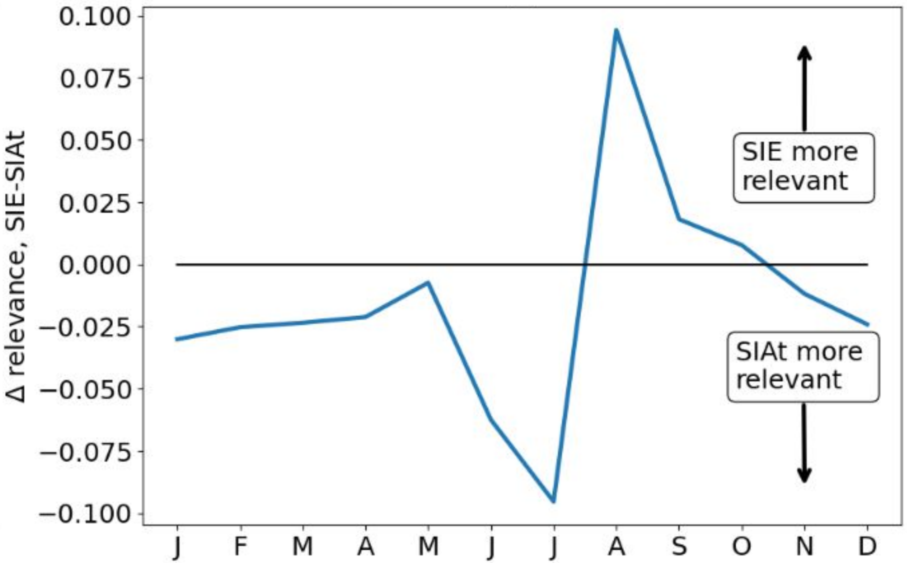
Hoffman et al. (*submitted, 2025*)

The *area of thick sea ice (SIAt)* is an important input predictor.

relevance



Δ relevance (SIE- SIAt)



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The *area of thick sea ice (SIAt)* is an important input predictor, particularly in the winter and spring months.

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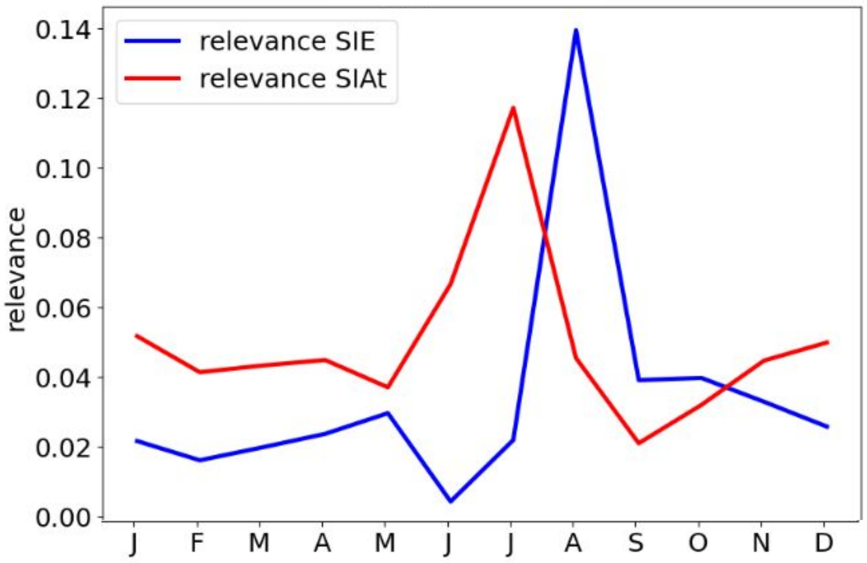

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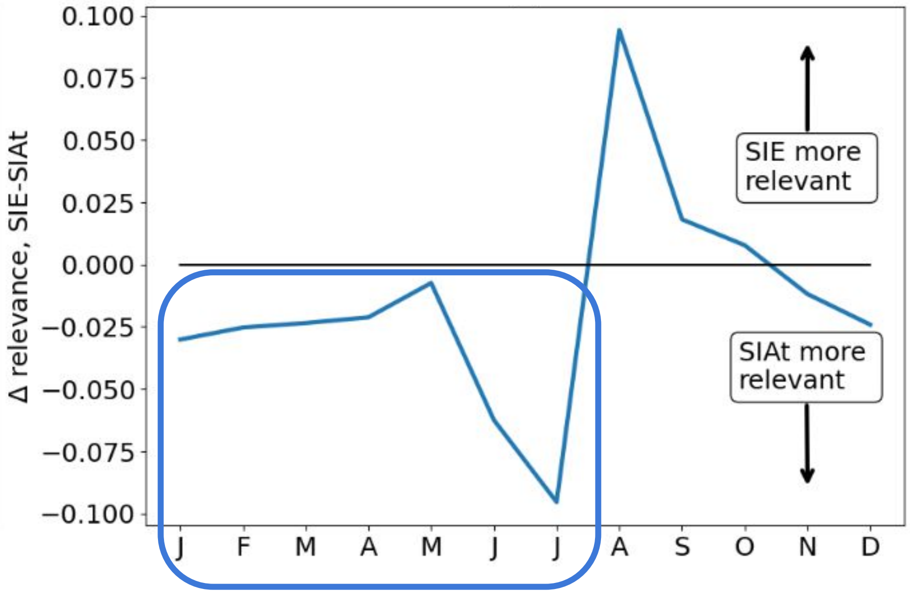

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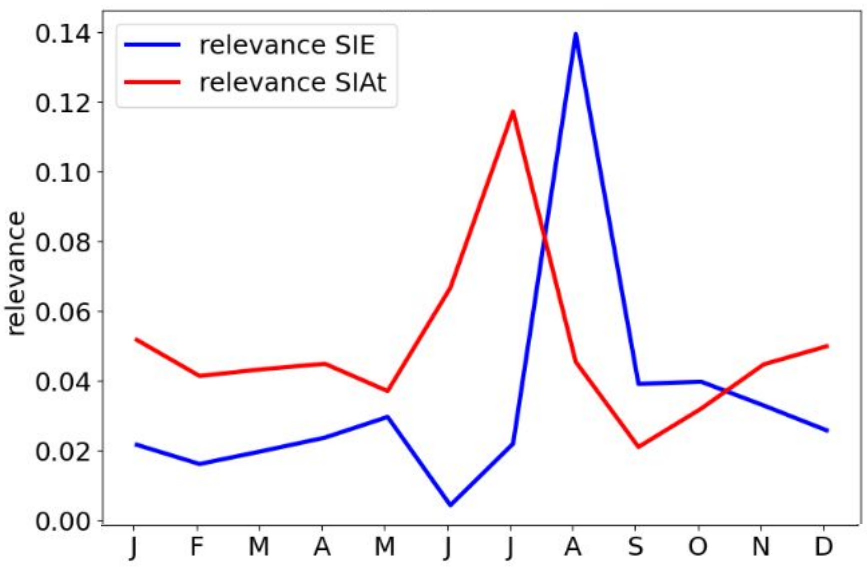
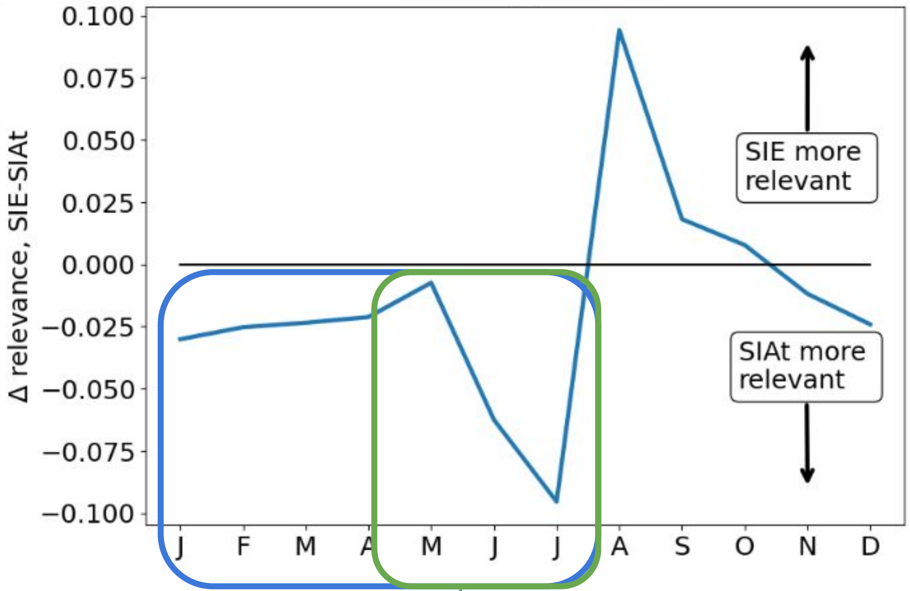


Δ relevance (SIE- SIAt)



Consistent with, e.g. Bushuk et al (2020)

Model Behavior

Hoffman et al. (*submitted, 2025*)

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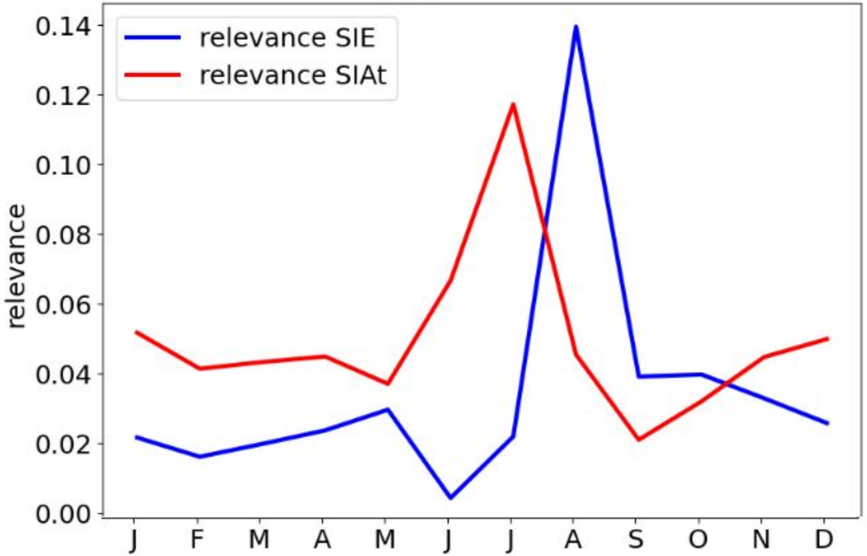
relevance



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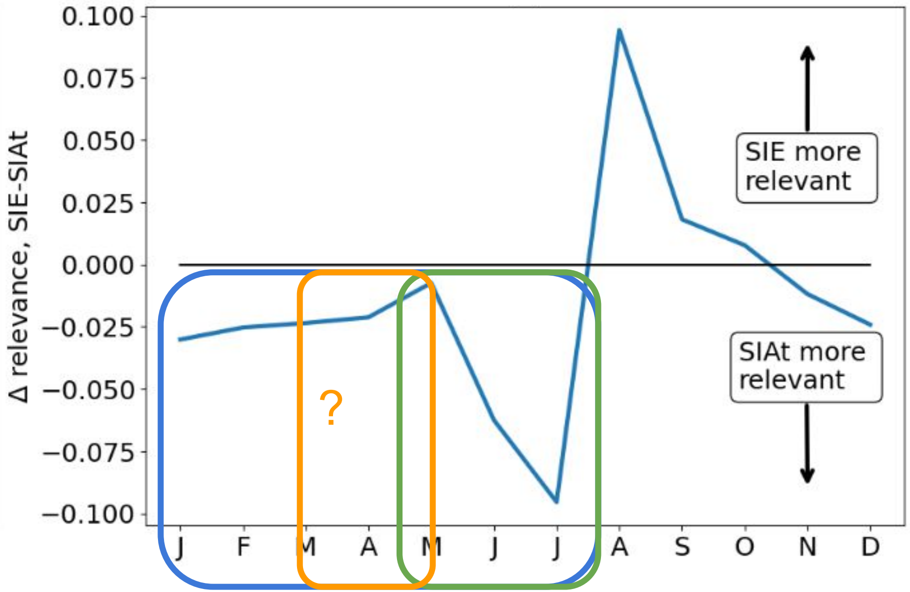
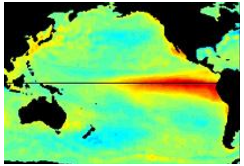


Model Behavior

Hoffman et al. (*submitted, 2025*)

Climate Regimes and Drivers



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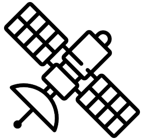

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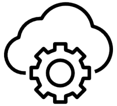

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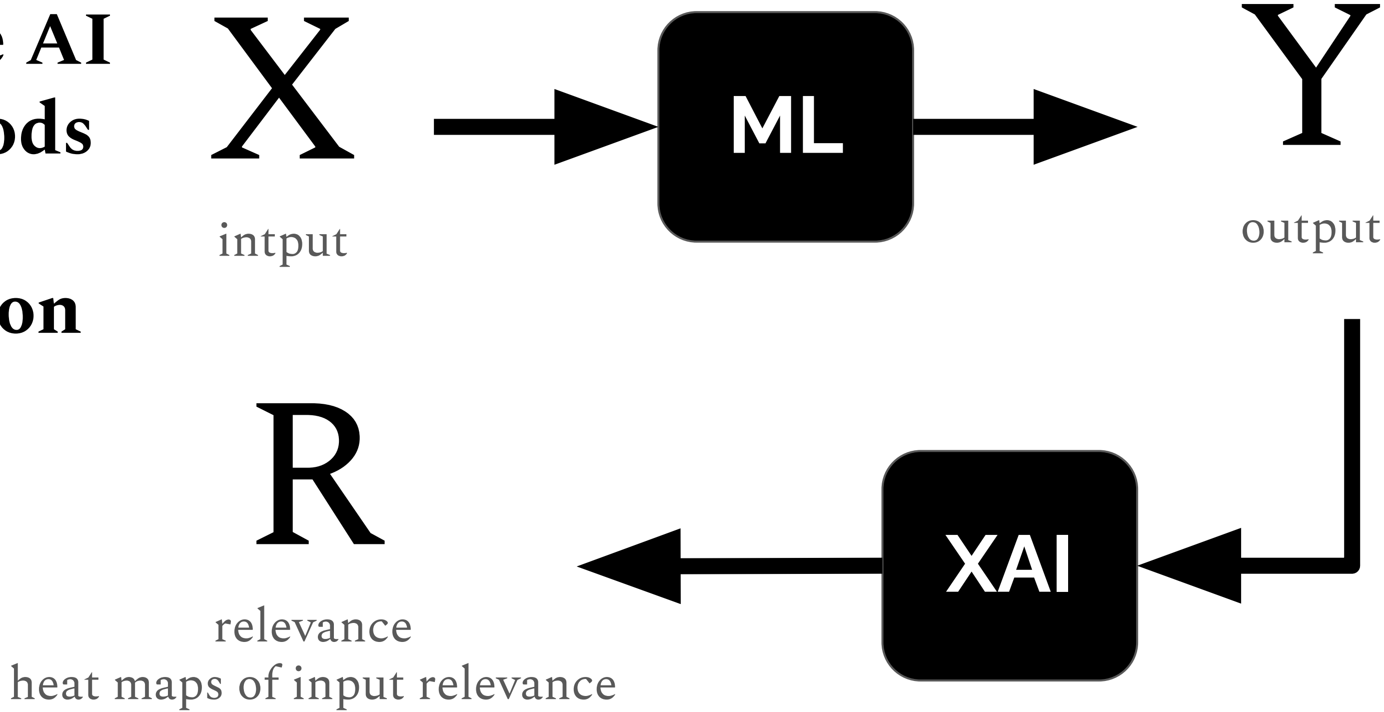
Climate Regimes and Drivers: XAI

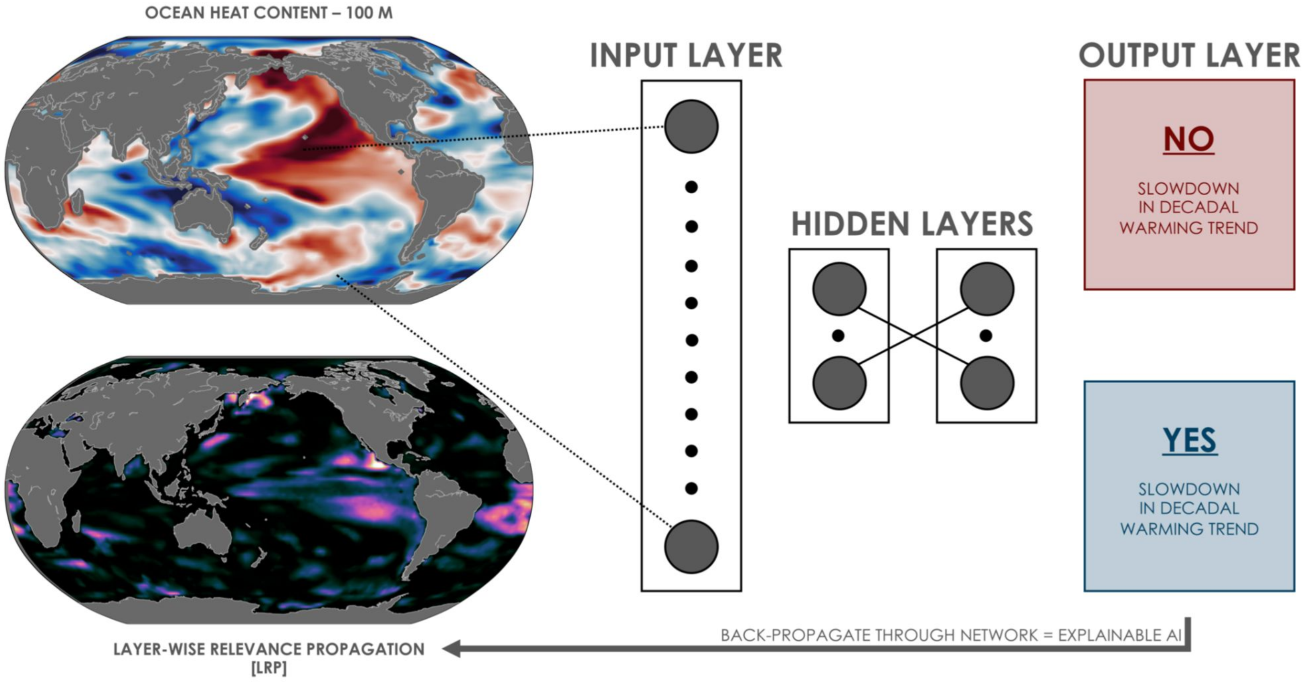
Explainable AI (XAI) methods show us an interpretation of what the model has learned.



Climate Regimes and Drivers: XAI

Labe & Barnes (2022)

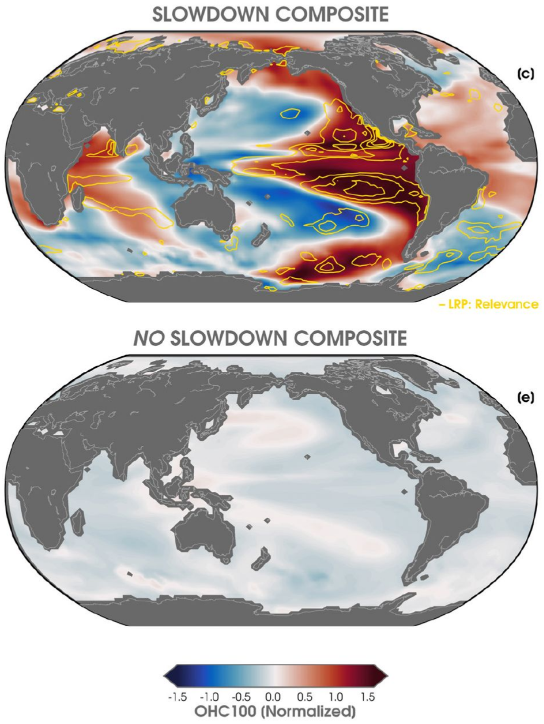
A neural network predicts slowdowns in decadal warming trends from maps of ocean heat content.

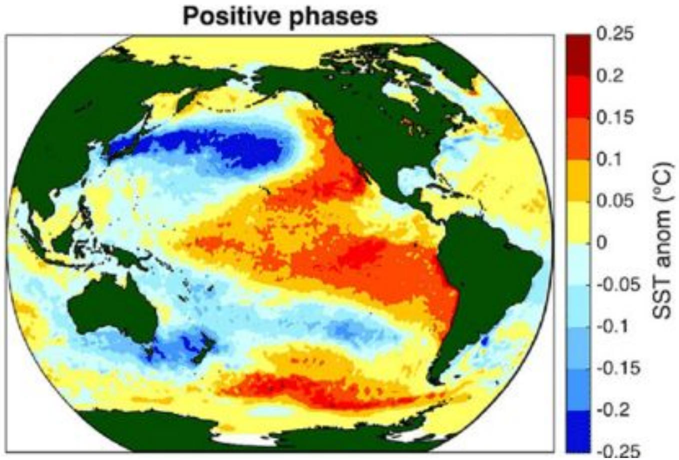


Climate Regimes and Drivers: XAI

Labe & Barnes (2022)

Composite maps of OHC where the NN correctly predicted a 'slowdown' resemble the positive phase of the IPO.





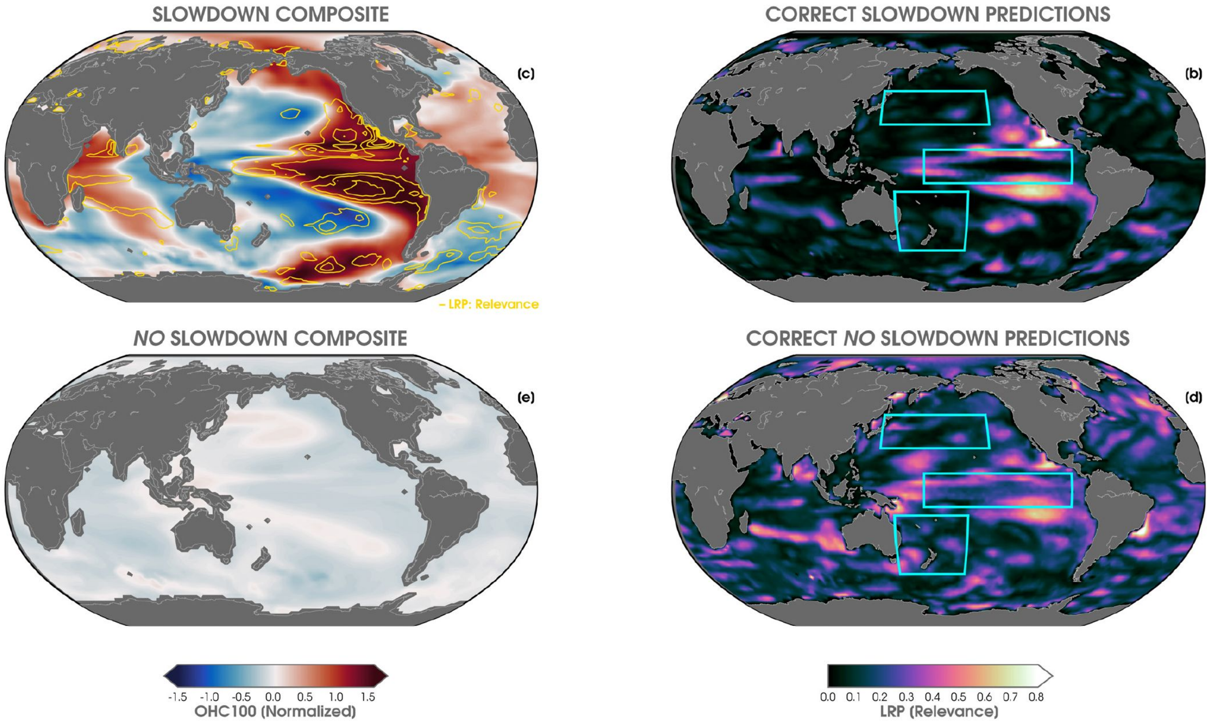
SST in the positive phase of the Interdecadal Pacific Oscillation (IPO)

Henley et al (2015)

Climate Regimes and Drivers: XAI

Labe & Barnes (2022)

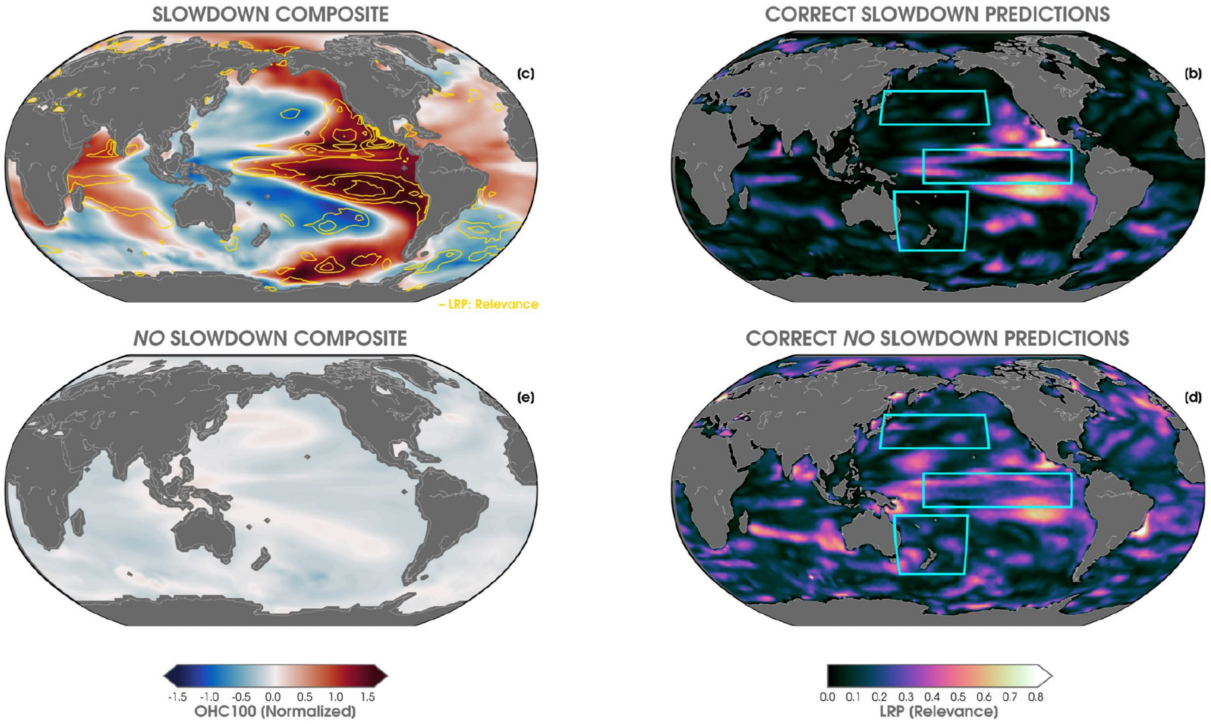
eXplainable AI shows where the input maps were relevant for predicting slowdown events.



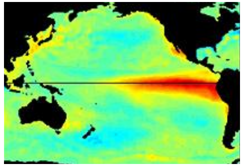
Climate Regimes and Drivers: XAI

Labe & Barnes (2022)

eXplainable AI provides these heatmaps at each timestep, so we can get an idea of the *evolution of the input relevance*.



Climate Regimes and Drivers



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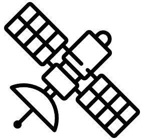

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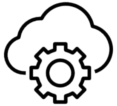

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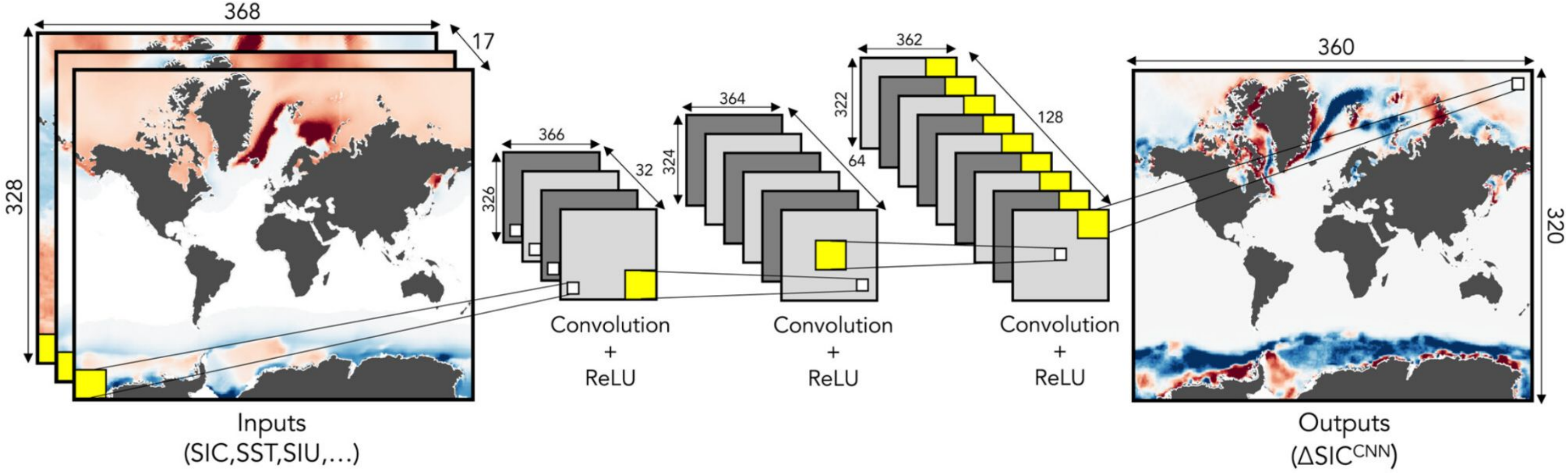
Earth System Modeling: Model Bias Correction

Gregory et al. (2023a,b)

A CNN trained to predict data assimilation errors in sea ice concentration (increments, ΔSIC) acts as a model parameterization to help correct model bias.

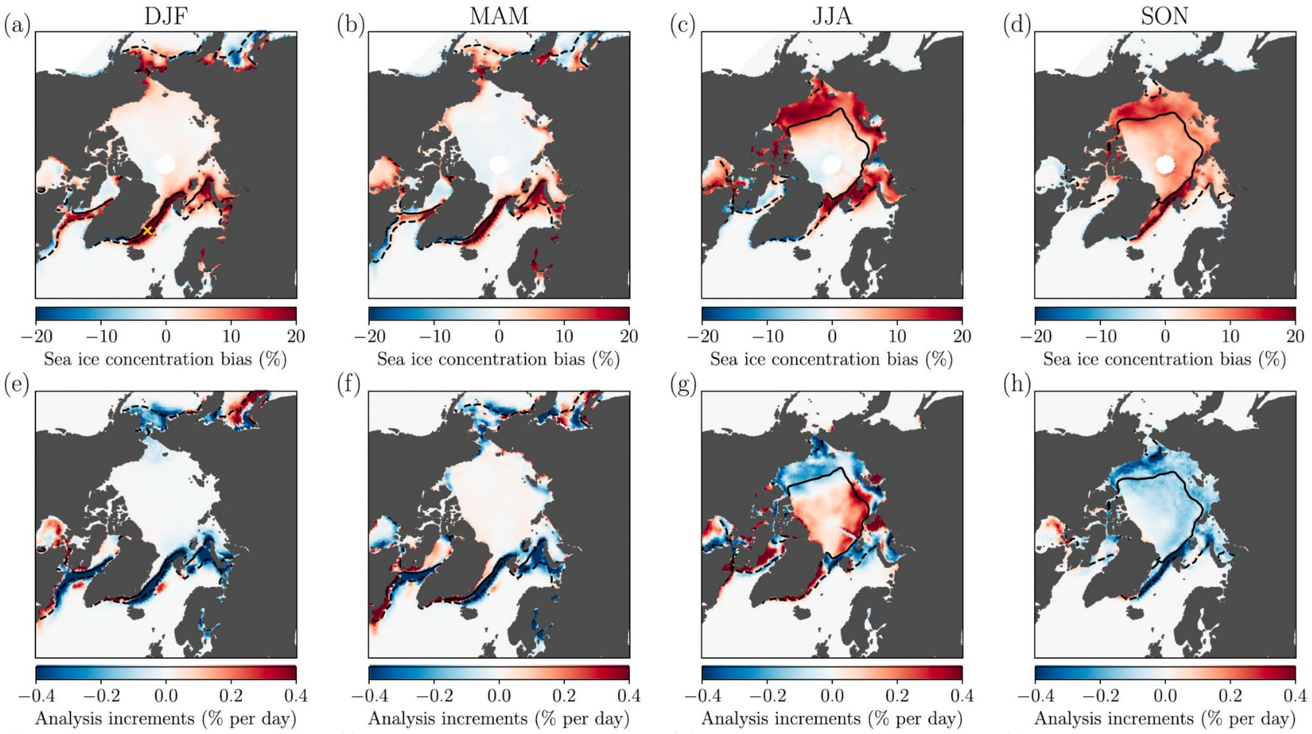
Model: SPEAR ice-ocean w/ JRA-55 atm. forcing



State variables & forecast tendencies

Earth System Modeling: Model Bias Correction

Gregory et al. (2023a,b)

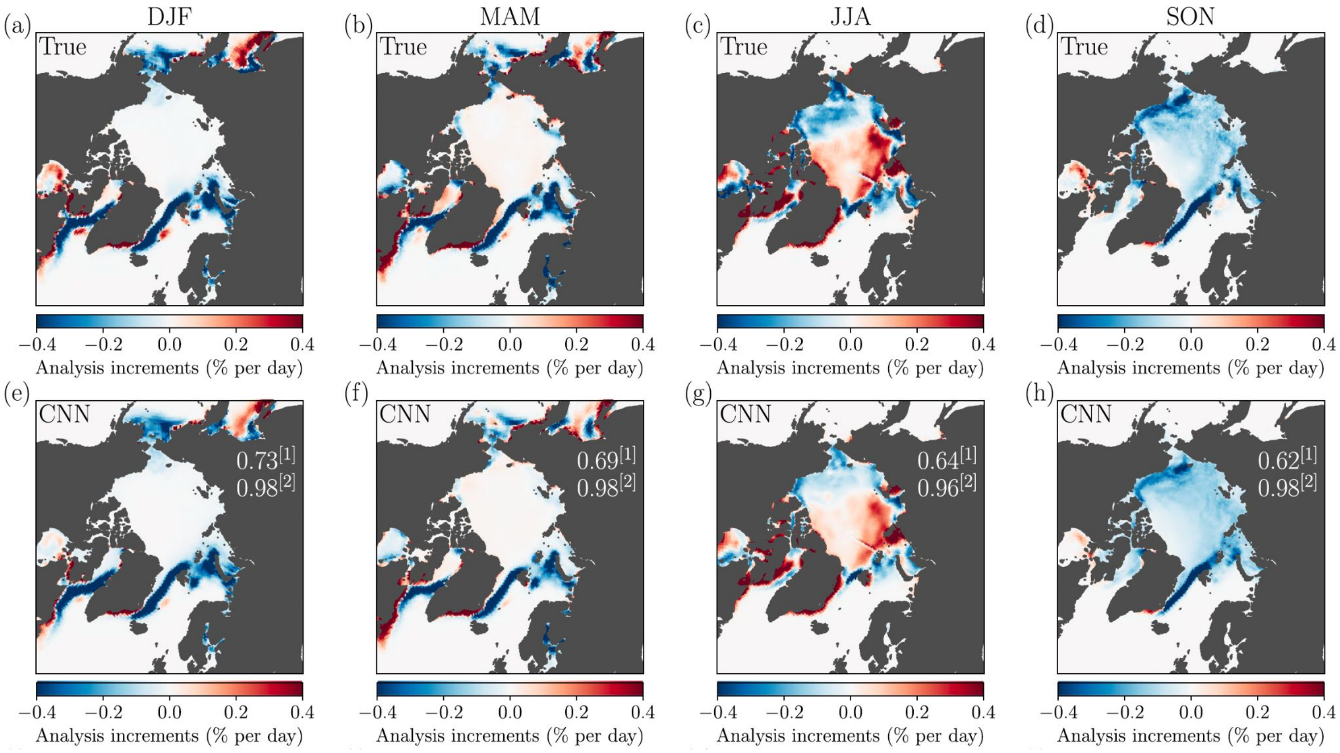


This work assumes that the analysis increments are representative of the SIC bias (FREE-NSIDC).

This is supported by the spatial and seasonal agreement between the free-running model bias and the SIC increment.

Earth System Modeling: Model Bias Correction

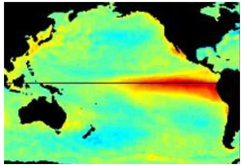
Gregory et al. (2023a,b)



[1] mean of the daily spatial pattern correlations between true and predicted increments.

[2] spatial correlations between climatologies of true and predicted (i.e. the maps)

Climate Regimes and Drivers



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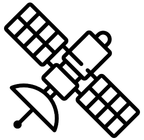

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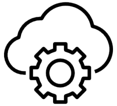

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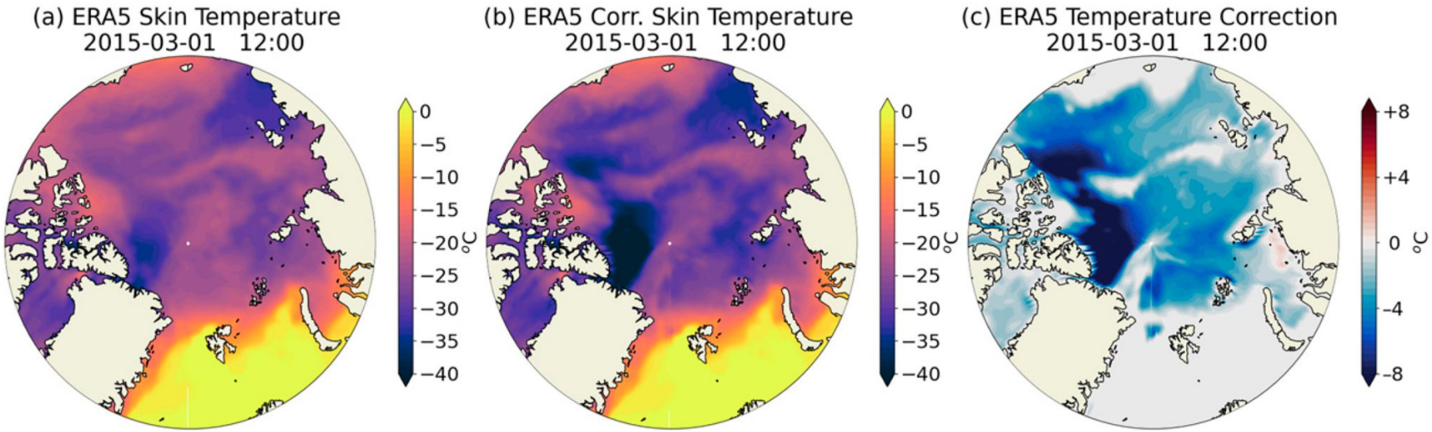
Data & Observations: Reanalysis Bias Correction

Zampieri et al. (2023)

A NN is trained to correct the Arctic temperature bias in reanalysis products, leading to a 27% and 7% bias reduction in ERA5 and JRA-55, respectively.



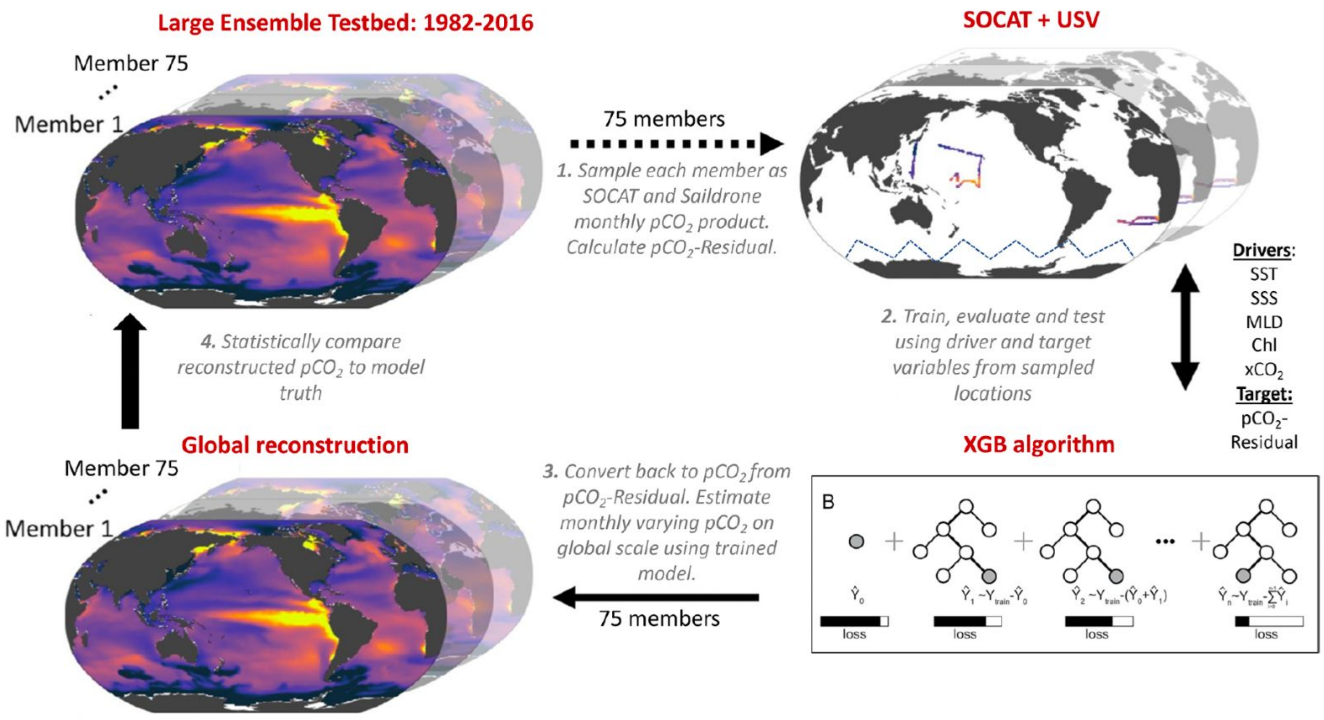
Inputs: Skin temperature (SKT)[ERA5], Surface downward longwave radiation (STRD)[ERA5], Sea ice thickness (SIT) [PIOMAS], Snow thickness on sea ice (SND) [SnowModel-LG]

Output: Skin temperature reanalysis bias (Δ ERA5, obs)

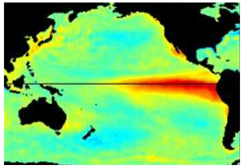
Data & Observations: Global pCO_2 reconstructions

Hemidal et al. (2023)

ML models trained on Earth System Models (ESMs) sub-sampled at the locations of observations from SOCAT + USV observations skillfully reconstruct global pCO_2.



Climate Regimes and Drivers



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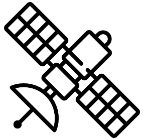

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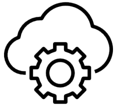

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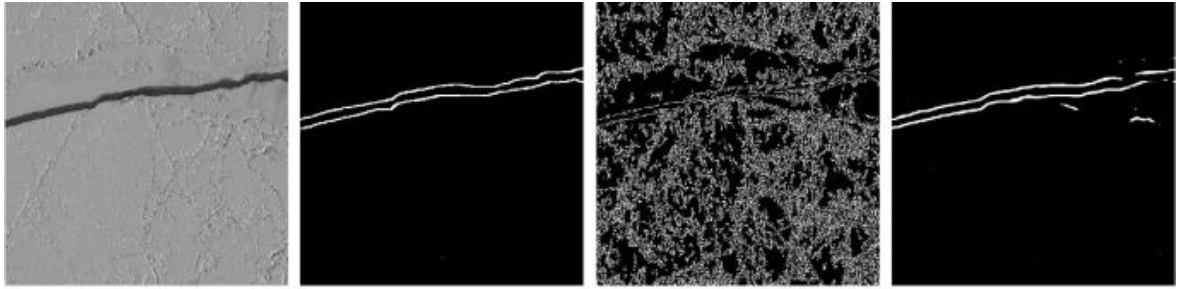
Feature Recognition: Lead Detection

Machine learning models are trained to classify pixels and detect leads using labeled datasets of satellite images (e.g. SAR).


(a) Image (b) GT (c) Canny [4] (d) U-Net [17]

Dulam et al. (2022)

Liang et al. (2022), Zhong et al. (2023), Kaltenborn et al. (2022), Hoffman et al (2021), Hoffman et al (2022), Kortikova et al. (2024), Qu et al. (2024)

Climate Regimes and Drivers

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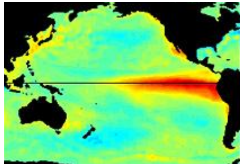

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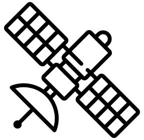

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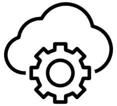

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AI4InSync

Optimizing Southern Ocean observing systems using AI.

lauren.hoffman@uclouvain.be
Lauren Hoffman, François Massonnet
Earth and Life Institute (ELI), Earth and Climate, Université catholique de Louvain

erc

ANTARCTICA INSYNC

AI4InSync

A collaborative workshop to support observation system design with machine learning methods.

Workshop Outcomes:

(a) Identify 1-3 scientific questions
(b) Identify potential working groups

InSync Webinar // 8-10am // 22 May

Session 1	Session 2	Session 3	Session 4
Motivation & Intro to ML	**ML Tutorial**	**State-of-the-Art: ML in Climate Science**	**Discussion:** Identify science questions Define potential working groups

AI4InSync

ML for observation system design.

Machine learning models can be useful tools for the development of polar observational networks.

(2026)
Current Climate Change Reports ~~(2019)~~ 5:334–344
https://doi.org/10.1007/s40641-019-00151-w

CLIMATE CHANGE AND SNOW/SEA ICE (PJ KUSHNER, SECTION EDITOR)

~~Climate Models~~ as Guidance for the Design of Observing Systems: the Case of Polar Climate and Sea Ice Prediction

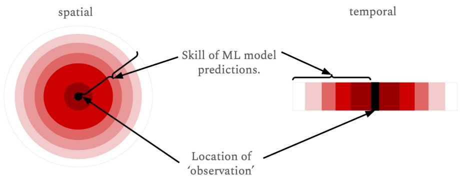
François Massonnet[1], Lauren Hoffman, Your Name Here

1. **Observing System Simulation Experiments (OSSE) and Quantitative Network Design (QND)**
2. Emulation of satellites and retrieval algorithms
3. Constraining long-term projections: identify observational gaps that, if filled, would allow uncertainty reduction in projections
4. Evaluation of observational products
5. **Strategic placement of in-situ sampling sites**

Potential ML Tasks

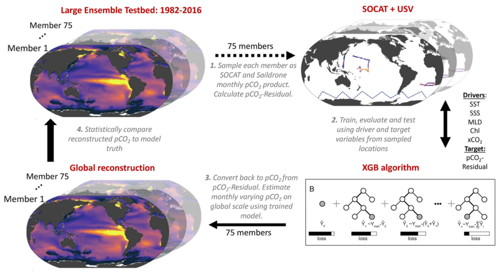
Can ML methods be used to optimize spatio-temporal frequency of measurements?

ML Interpolation



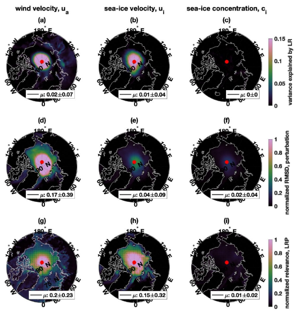
How far away is a ML model able to skillfully predict?

Global reconstructions



How well does knowledge of *[variable]* at sampled locations allow for global reconstruction of *[variable]*?

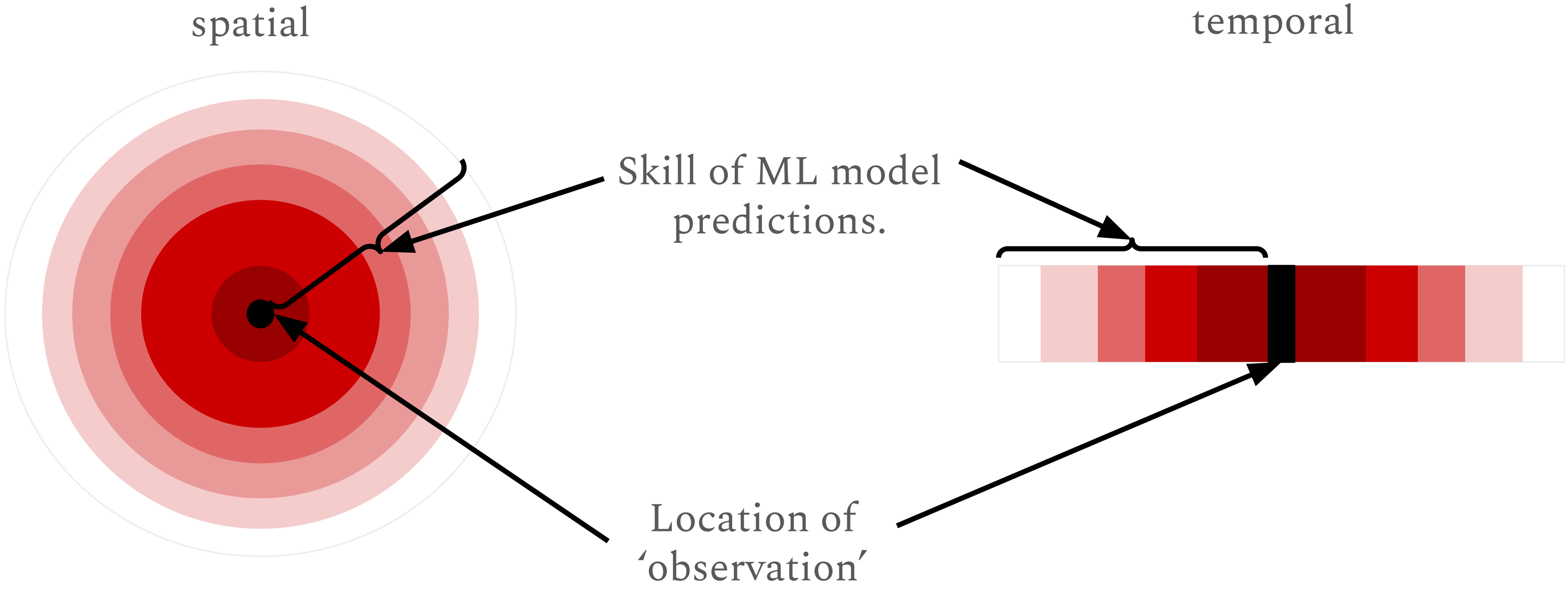
Explainable AI



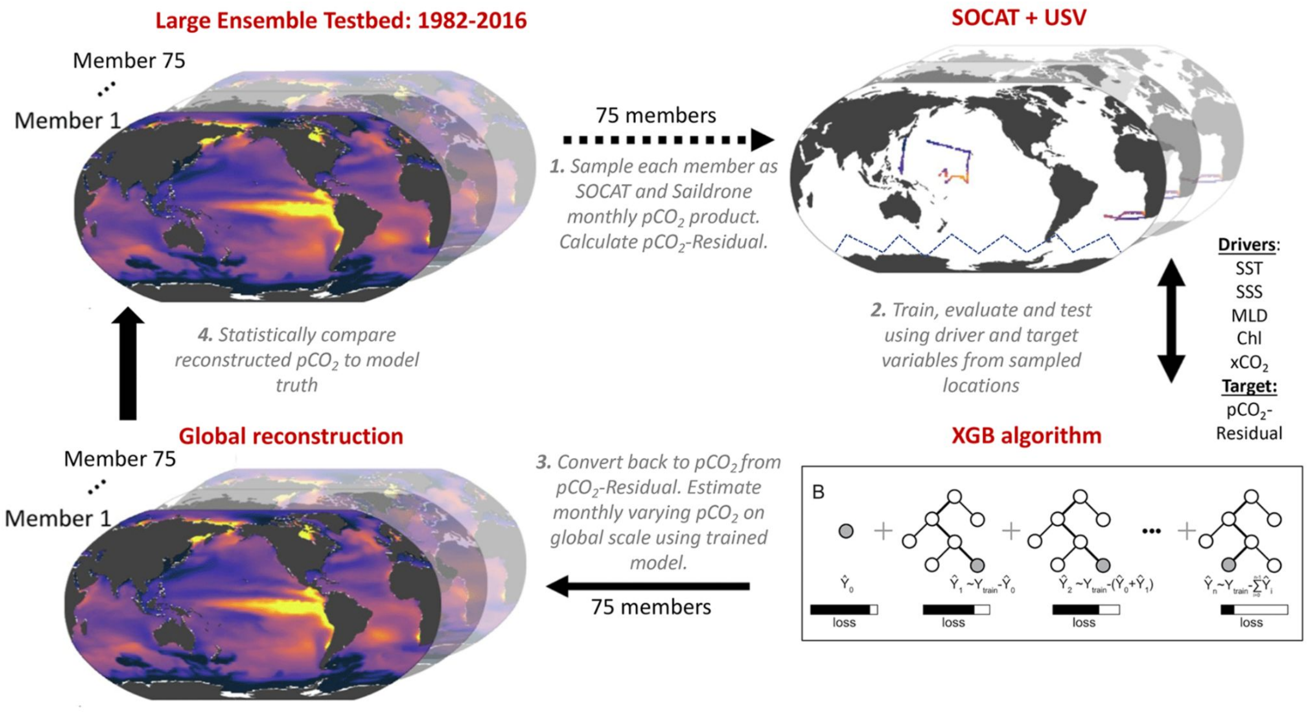
Where are *[inputs]* relevant for skillfully predicting outputs?

ML Interpolation: Use model outputs or satellite data to train ML model to predict *[variable]* at distances surrounding a centered location or time.

How far away is the model able to skillfully predict?



Global reconstructions: How well does knowledge of pCO2 at sampled locations allow for global reconstruction of pCO2?



Use LE testbed to train ML to learn relationship between [inputs] and [target variable] at [observed locations].

Use spatial maps of inputs to reconstruct global map of output.

Inputs: available satellite measurements.

Target variable: in situ observed.

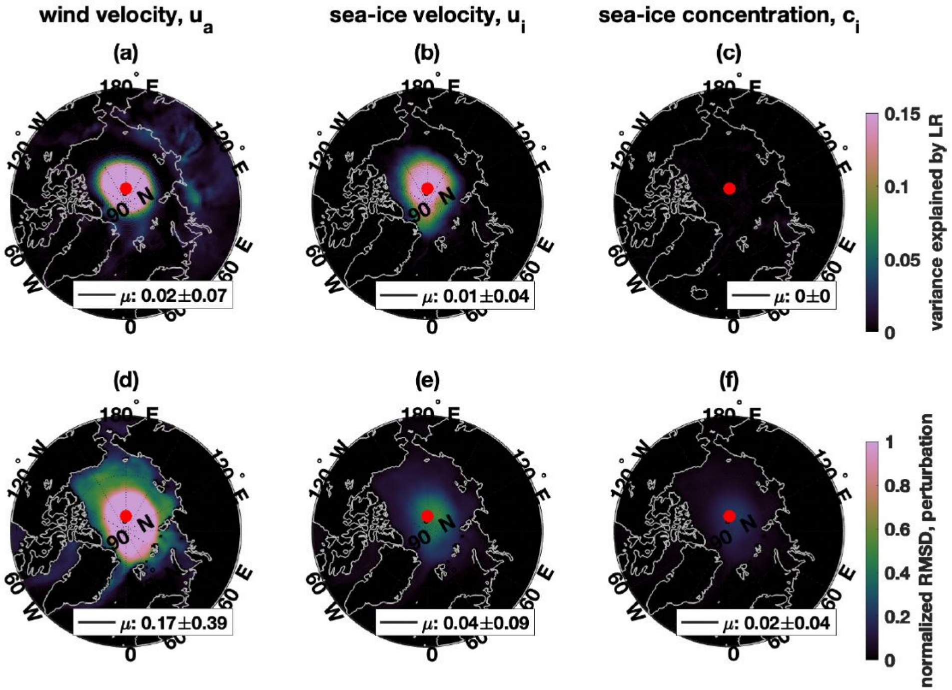
Observed locations: which spatio-temporal observation frequency provides best global reconstruction?

Assessing improvements in global ocean *p*CO2machine learning reconstructions with Southern Ocean autonomous sampling. Hemidal, McKinley, Sutton, Ray, Gloege (2024). https://doi.org/10.5194/bg-21-2159-2024

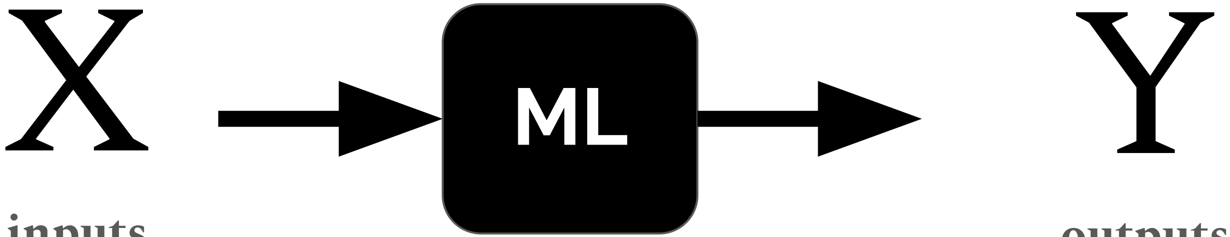
Explainable AI methods show where inputs are *relevant* for predicting outputs.

Variance explained by linear regression (LR)

Perturbation



Can ML be used to generate satellite-derived under-ice eddies?



inputs
surface variables:
SST, SSS, SSH, SIC,
wind, SIT, sea ice
velocity

model
probabilistic
regression,
outputs are μ and σ

outputs
under-ice eddies (MIZ),
i.e. EKE, vorticity

I. **Training: NEMO-SI3, 1/12º resolution**
II. **Satellite-derived ‘product’: trained model makes prediction of under-ice eddies from inputs of surface variables from satellite measurements.**
III. **Regions of high uncertainty in model predictions are used to inform where further observations could be useful.**
IV. **Transfer learning: fine-tune model weights / learned parameters with obs**

Questions for Breakout

What are your needs for an observing system?
- Variables?
- Spatio-temporal scales?

How do you define the value of an observing system?
What objective function would you optimize?
- Reducing mean state error?
- Constraining long-term trends?
- Improve forecast skills?
- Capture extreme events?
- Uncertainty reduction?

How could optimizing for one metric harm another?

Bringing it together: What are 1-2 scientific questions you'd like to have answered with the InSync campaign?